User manual

How the application works, screen by screen. The interface is in English, so every control appears here under the literal label you see on screen, explained beside it. The version history is not in this manual: it is in the `CHANGELOG.md` file that ships with the application, and the project's record to date is in `docs\HISTORY.html`.

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This tool supports the design process. It does not replace engineering approval, a site survey, cable route validation or physical inspection. — This line sits at the foot of every screen and on every exported PDF, and it is not decoration: no number here replaces engineering approval, a site survey, cable route validation or physical inspection.

Concepts and acronyms

These terms appear on nearly every screen. Understanding the difference between a *room* and an *occurrence*, and between *extracted* and *confirmed*, is what separates using the tool from trusting it.

TERM	MEANING
NER	Network Equipment Room. The room that serves an area's network points. It is **critical** or **non-critical** — and in this application that classification only counts once a person confirms it. **Only a NER serves a space.**
TER	Telecommunications Equipment Room. Read off the drawing and drawn, it can be placed and moved, but it **never serves a space** and gets no reach circle — the TER↔NER relationship is a design rule this tool does not invent.
IDF / MDF · COW · DIVA node	Other markers the drawing carries or you place: intermediate/main distribution frame, a temporary cabin (*cell on wheels*) and the DIVA rack. **Visual only**: they appear on the plan and on the exported PDF, and never serve any space.
TEAP	The workbook stating technology equipment and service demand per location. It is the source of the question “how many network ports does this space need?”
VDB	The venue's drawing set (PDF), used to identify spaces, functional areas, TERs and existing NERs. It is the source of the question “where is this space?”
TVDB	The later document carrying containment, pathways, routes, shafts and risers. **Not read by this version** — see what the app does not do.

TERM	MEANING
Space	A named room or area. It exists in two places: on the drawing (a *drawing space*, with a page and a position) and in the TEAP (a *TEAP space*, with a port demand). The TEAP × drawing step is what ties one to the other.
As-Is	The name of the saved coverage analysis: the drawing as it is, with whatever you have placed on it. The Demand per space screen and the reports read who serves what from it.

Room and occurrence

A design book draws the same room several times: once on the level key plan, again on a zone detail, again on a services overlay. The application keeps the two apart:

- **Canonical NER** — the physical room. It has a name, a classification, a capacity, a position, a lock and a confirmation. Its internal identifier and its code *never* change, not even when you rename it.
- **Occurrence (marker)** — where that room appears in a document: file, page, coordinate, the text the extractor read and the detection confidence.

A room drawn on four sheets is *one* row on the NERs screen with four markers — not four NERs. Without that separation the room count is inflated, and every number downstream inherits the error.

Four kinds of data, never mixed

KIND	MEANS
extracted	what a document said — the label read off the sheet, the printed scale, the TEAP line.
confirmed	what a person signed off — a page's scale, a TEAP × drawing match, a NER's criticality.
estimated	what the tool inferred to keep going — a service point at the centre of a space, a port count assumed from the category.
decided by you	what you chose by hand — the room serving a space, the drawing room that corresponds to a TEAP line. It outranks the automatic answer, lands in the audit log, and is still checked against distance and capacity.

Every row, every match and every result carries which of these it is. The exports say so row by row (`Needs validation`); the screens say so with pills, rings and notes.

The workflow

The left-hand navigation *is* the workflow, in order. Every step is disabled until a project is open.

The navigation **folds**: `«` at its top right shrinks it to a narrow rail of step numbers (`1` to `9` , with the name of each step when the pointer rests on it; `▤` is Projects and `⚙` Account & security), and `»` at the top of the rail opens it again. The drawing takes the width it leaves — useful on a laptop screen. The browser remembers the choice.

The application opens in the browser at `http://localhost:8000`, or from the **NER Analyzer** shortcut `setup.bat` puts on the desktop. It works in **Google Chrome, Microsoft Edge and Mozilla Firefox**, in a current version; Safari and Internet Explorer are not supported. The installer, the shortcut and every message use `localhost`: on a corporate machine Microsoft Edge may open `http://127.0.0.1:8000` — the same application — in *Internet Explorer mode*, where the page stays blank.

STEP	TAB	WHAT HAPPENS
—	Projects	Create, open, share and delete projects. A project is one venue.
1	Import	Upload the VDB PDF and the TEAP workbook, map its columns, pick the venue and set the distance rules.
2	Scale	Tick the key plans (the sheets that only illustrate), confirm the scale of every page — definitive results stay blocked until you do — then **place the NERs from the drawing**: the team's annotations, the coded rooms (`TEC.010` / `TEC.020`, or `TEC.003.01` / `.02` on older books), the NER labels, the NER block symbols — named in sequence for you to rename.
3	Drawing viewer	See the real plan; create, move, rename and delete markers (NER, TER, IDF, COW, DIVA node) and Wi-Fi points; trace routes and register shafts; paint the coverage over the plan.
4	NERs	One list, one row per NER — inherited from the drawing or drawn by you: name, origin, criticality, indoor / outdoor, capacity, lock, confirmation, removal; duplicates and history; the naming panel beside it.
5	TEAP × drawing	Match TEAP rows to drawing spaces, in demand order, and confirm them one by one. This is where the work actually happens.
6	Demand per space	Run the analysis; port demand per space, the NER serving each one, the distance, the status and the load per NER.
7	Wi-Fi	**Place Wi-Fi from the TEAP** — one Wi-Fi point beside the room of each space the TEAP says has wireless, tied to its space — once the rooms are settled on TEAP × drawing; the list of Wi-Fi points, with the NER serving each one, the distance and the status.
8	Exceptions	Every TEAP space that still needs you — pending, out of the limit, near the limit or not measured — with the NER changed on the row itself.
9	Reports	Marked-up PDF, composite map PDF, analysis workbook and CSV.
↗	Composite map	An option, built from the Reports screen when you ask for it: the detail sheets of a level on one map, in a window of its own, where NERs are moved, deleted and reclassified with their reach in view.
—	Account & security	Your password, the accounts on this installation (if you are an administrator), the open project's audit log, and the statement of where the data lives. It sits below a divider because it is not a step in the study.

The top bar, on every screen

- The project name, or `No project selected`; beside it, `venue VEE` when the project has a venue code.
- A green `Scale confirmed` pill, or an amber `{n} page(s) awaiting scale` (with no drawing yet, `0 page(s) awaiting scale`).

- A red *server not answering — reconnecting* pill while the server is out (see Running).
- The limits in force: `limits 90 m / 70 m · slack 1×`.
- An amber pill *the application was updated — press Ctrl+Shift+R to load the current interface* appears **only** when the browser is still holding an interface from another version. Press `Ctrl`+`Shift`+`R` and it goes. The project that was open **reopens by itself** after any page reload (if it still exists and your account can still see it).
- Your name — it is a button and opens Account & security (for an administrator the tab reads `Accounts & security`); it shows `· admin` if you are an administrator and *change your password* when a change is required.
- `PT-BR` / `EN` switches the interface language (see Language), and `Sign out` ends the session.

In the sidebar, under the title, is the application's **version** (`version 5.2.2`) and, at the foot, the author credit and the licence.

Notices

What an action did is said in a **notice** that appears at the top of the window, over the page, green when it worked and red when it did not. It does not push the screen down — in the Drawing viewer the sheet stays exactly where it was under the pointer — and it goes by itself after a few seconds; a click closes it sooner.

The progress bar

Long operations — importing a 75-page PDF, matching thousands of rows, running the analysis, writing the marked-up PDF — run in the background and appear in a bar reading `Working` / `Finished` / `Failed` / `Cancelled`, with the current stage in words (*reading page 12 of 34*), the percentage, and the buttons `Cancel` (while running) and `Dismiss` (afterwards). The bar **floats at the bottom right** of the window, over the page, so it never pushes the screen down or up as it opens and closes. The rest of the application stays usable meanwhile. A finished operation closes its bar by itself after a few seconds — what it found is in the notice at the top. When an operation fails, the message stays in the bar until you dismiss it.

If a screen fails

A screen that fails to draw shows the failure **inside itself**, with the message, a button to try again and one to reload — and the rest of the application stays up. Moving to another tab clears it.

PROJECTS

Projects

Create, open, share, hand over and delete projects. A project is one venue: a drawing set, a TEAP import, and the decisions made about them.

Card “New project”

CONTROL	WHAT IT DOES
`Project name` TEXT	The project name. `Create` does nothing if it is blank, and spaces at either end are dropped. The name must be unique on the installation, whatever the case — *VAA test* and *Ala Test* are the same name: a repeated one is refused with *a project with that name already exists*.
`Venue code (optional)` TEXT	The venue code, for example `VEE`. **Always fill it in** — it shows in the top bar and step 1 uses it to suggest the right TEAP filter.
`Create` BUTTON	Creates the project, selects it and jumps straight to `Import`.

Projects table

COLUMN	SHOWS OR CONTROLS
`Name · Venue`	Name and venue code.
`Owner`	`you` on yours; on other people's, the owner's name. An administrator sees everyone's projects; everyone else sees their own and those shared with them.
`Visible to` SELECT	`only me` (private) or `accounts on this machine` (shared). Sharing grants the other accounts on this installation **reading**; it publishes nothing anywhere and grants no editing. Only the owner or an administrator changes it; the recipient sees *shared with you · read only*.
`Drawings · TEAP files · Scenarios`	How many PDFs and TEAP workbooks were uploaded to the project (every uploaded workbook counts, even superseded or never imported) and whether an analysis has been saved (`1` ; every `Run analysis` replaces the previous one).
`Open`	Opens the project and jumps to `Import`.
`Export`	Writes the **whole project into one package** — a `.zip` named `NER-project_<name>_<date-time>.zip` — and the browser downloads it. The button reads `Exporting…` while the books are copied in (the progress bar on the import card shows the stage). The message at the end gives the size and how many documents went in. Anyone who can read the project can export it — including a project shared with you for reading, which is how you take **your own copy** of it. See The project package below.
`Delete`	Asks *Delete “name” and everything in it?* first. Disabled on projects that are not yours — except for an administrator, who can delete any project.

CAREFUL

`Delete` removes the whole project — drawings, TEAP, matching, NER list, analyses and audit log, and the uploaded files and the exports on disk. Who deleted it, when, and what it held (its name, books and TEAP files) stays in the installation's own log — *Accounts & security* → account events. Its number is never given to another project. (At every start the application also looks for files no project points at — an upload that failed or was cancelled half-way — and moves them to `data\set-aside\<date-time>\` with a note, `WHAT-IS-THIS.txt`; it never deletes them. Delete that folder to free the space.) There is no recycle bin. NERs removed inside a project can be restored; a deleted project cannot. Take a backup first (`backup.bat`, see Running).

Card “Import a project package” — handing a project to another computer

When a project has to change hands — you stop, or another VTCM takes the venue over — the next person does not start from zero. `Export` on the projects table writes the project as it stands; on the other computer this card brings it in.

CONTROL	WHAT IT DOES
`Name for the imported project (optional)` TEXT	The name the project will have here. Left blank, it keeps the name it had; if that name is already taken on this installation, `(imported <date>)` is added to it.
`Package (.zip)` FILE	Choosing the package starts the import, in the background, with the progress bar below. When it finishes the new project is opened, and the message says who exported it, when, from which version — and what the analysis gave here.

What travels: the design books and the TEAP workbooks as they were imported, and every row of the project — pages, scales and key-plan ticks, the rooms read off the books, the NER list with names, criticality, indoor/outdoor, positions, locks and removed rows, the TEAP spaces with your ties, rejections and chosen NERs, the matches and their undo, the same-space links, the Wi-Fi points, the illustrative areas and references, cable routes and shafts, the project rules, the aliases, and the project's audit log. The column-mapping templates go along too (a template whose name already exists here is left as it is).

What does not travel: accounts and passwords (the person who imports becomes the owner, and the project is private until they share it), the background jobs, and the saved analysis itself — which is **run again** on import. The analysis is deterministic: the same books, TEAP and decisions give the same answer, and the message says so — *came out the same*, or, if the exported analysis was already out of date, that it was run on the current data. If it ever says the two differ, open Demand per space before carrying on.

A COPY, NOT A LINK

Importing creates a **new** project with new numbers; nothing existing is changed, and the original stays where it was. From then on the two copies are independent: there is no merge and no synchronisation. Agree who carries on — whoever handed the project over should stop working on their copy, or the two will drift apart with no way of putting them back together. A package can be imported only by the same version of the app or a newer one; a package from a newer version is refused with the advice to update.

THE PACKAGE HOLDS THE VENUE'S DOCUMENTS

It carries every design book of the project in full. It is not encrypted. Move it only through the channels your organisation approves for those documents — never a personal e-mail or a public link — and delete it once the other side has imported it. The application keeps only the newest package of each project in its `exports` folder. Every package is checked before anything is written: only the files its manifest lists are read, each against the SHA-256 recorded on export, so a package damaged or altered in transit is refused whole — *… does not match the copy that was exported (damaged or altered in transit). Nothing was imported.*

WHY SHARING DOES NOT GRANT EDITING

The application has no concurrency control. Two people confirming different matches on the same project would silently overwrite each other. Until that is solved properly, writing stays with the project's owner (and the administrators). The recipient of a shared project can open it and move through every screen; any change — building a report included — is refused with *this project is shared with you for reading.…*.

STEP 1

Import

Three numbered cards: the drawings, the workbook, and the project rules.

1 · Drawings (PDF)

Choose one book or several at once — Ctrl or Shift in the file dialog. The import starts at once, in the background, with the progress bar naming the page it is on; several books are read **one after the other**, each to its end, and a list under the bar says of each one *waiting*, *reading*, *imported* (with its pages and spaces) or *refused* (with the reason). A book refused does not stop the others. The book is read by a second Python process started for the job (Task Manager shows it beside the server while the import runs); it ends with the import and gives its memory back, which the server itself would not. When it finishes it reports how many pages and spaces were read, how many NER annotations and NER rooms the book states (placed later, on the Scale screen) — and, if any, how many pages need OCR. From every page the importer reads the **title block** (drawing number, sheet title and level, when the title states one), the printed scales, the space labels and the room markers. On a

sheet whose title block was read, the level is the one its *DRAWING TITLE* states (`… - LEVEL 01`) or none — a bare three-digit figure printed large on the sheet (a stand's seat count) is never taken for a level.

The table lists: `File`, `Pages`, `Spaces`, `Markers`, `OCR needed` (the scanned pages' numbers, or `none`) and, on each row, `Remove`.

> THE SAME BOOK ONCE
>
> A book is refused, and nothing of it kept, when **the same file** is already in the project — whatever it is called: *'…' is already in this project (the same file, imported as '…'). Nothing was imported.* — and when **a book with the same name** but other content is: that is almost always a new revision, and two revisions side by side draw every room twice: *a book named '…' is already in this project, with other content — a new revision? Remove the old one on the Import screen first, then import this one.* The same file picked twice in one selection is dropped before anything is sent. A project that already holds the same file twice shows *same file as … — remove one of them* under the second copy.

Remove — taking a book out

`Remove` first counts what goes with the book and asks: its pages, rooms and markers; the NERs drawn **only** on it, by name; how many NERs are also drawn on another book; how many TEAP lines you tied to one of its rooms; and whether it is the same file as another book. Confirmed:

- its pages, its rooms and everything drawn on it — markers, Wi-Fi points, illustrative areas, references, cable routes and shafts — go, and so does the stored PDF;
- a NER also drawn on another book **stays**, placed where that other book draws it;
- a NER drawn only on this book leaves the list, removed with the reason *its book … was removed from the project* (it shows among the removed NERs; `Restore` brings back the row, with no position, since its sheet is gone);
- a TEAP line you tied to one of its rooms is untied, and the answer says how many — tie them again on the books that remain;
- the matching runs again on the books that remain, so the TEAP lines find their rooms there.

It cannot be undone; run `Run analysis` afterwards. `Remove` waits while a background task of the project is running (*a task is still running on this project…*).

SEVERAL SERIES OF ONE BOOK IN ONE PROJECT

A venue design book comes in series — 300 (floor plans, a whole level per sheet), 400 (enlarged plans), 600 (seating bowls), 700 (field of play) — each a PDF of its own whose pages start at 1. **Upload the series you want into the same project**, one after the other: a space that only one series shows (at VAA the press tribune and the commentary positions are on the 600 series and nowhere on the 400) can then be crossed with the TEAP. Each sheet is known by its **book and its page**, never by the page number alone: the viewer, the lists, the exports and the analysis never take page 2 of one book for page 2 of another. The screens name a sheet by its series and page — `400 page 6 · VAA-411B3-VBV-O`; the workbook by its drawing number — `VAA-411B3-VBV-O (400 p.6)`.

- **One NER, one row, whatever the series.** A room printed with the same numbered code on two series (`TEC.010-007` on the 400 and on the 600) is one NER with a marker on each sheet. Placed by hand on another series, it is the same question as on another page: *is it the same NER?*
- **One space, one row.** The TEAP line counts once. The same space code on another sheet, on the same level read off both title blocks, is linked to it as *the same space*; with a level unread on either sheet it is a question for you; on another level, nothing.
- **Distances are measured within one sheet** — the space and the NER both on it, at that sheet's scale. A space on a 600 sheet with no NER drawn there is *pending* until the NER is drawn on that sheet too (or chosen under `Served by`).
- **The marked-up PDF** is one file with every book, in the order they were uploaded.

ONE DRAWING VERSION PER PROJECT

A file that cannot be read as a PDF — damaged, password-protected, or not a PDF under a `.pdf` name — is refused with *'...' could not be read as a PDF — the file is damaged, protected or not a PDF. Nothing was imported.*, and nothing of it is kept. A copy cut short in transit, which opens with no page at all, is refused the same way (*... it opens with no page (damaged, truncated or not a drawing)*).

Uploaded the wrong book? `Remove` takes it out (above). **Do not upload a revised VDB into a project that already has the same series.** A file with the same name is refused; a revision under a new name is not caught, and the application takes every book of a project as part of the same drawing set, so the old and the new 400 series side by side are two sets of the same rooms — the analysis would measure from a marker on one version to a space on the other. It does not error. The numbers stay plausible and are wrong.

When the VNI team returns the revised VDB, either `Remove` the old series and import the new one in the same project — the NERs drawn only on the old book leave the list, and the rooms are matched again — or create a **new project** for it and carry the previous analysis workbook's *NERs and ports* sheet (Reports) across for reference. One project per *venue* is right; one project per *version of the same venue* is the rule for as long as the application does not version drawings.

A TEN-SECOND CHECK

After uploading the VDB, open the Drawing viewer on a page with named rooms. The grey space dots should land **on** the rooms in the drawing, and every NER marker on the label the drawing prints. If they are displaced in a systematic way, the problem is the document (an unusual page rotation), not the data — worth raising before moving on to scale and matching.

HOW A SPACE IS RECOGNISED

The importer first looks for the **space code** printed beside the room — the accommodation-matrix (AAM) naming: a functional area and a number, `ACR.001`, `PRS.006.02`, `BRS.052.02-001` — and builds the space from it: code, name on the neighbouring lines, area. That is the case of the books that follow the TEAP taxonomy. **On a sheet that prints no code at all** — every room labelled by name alone, over two or three stacked lines — a name label becomes a space **only when it reads as a known space name**: one of the names of an *Accommodation Area Matrix* (AAM) list, when the copy carries one in its private reference folder (`backend\app\reference_private\aam_names.txt`: `Venue Accreditation Office (VAO)`, `Broadcast Mixed Zone`, `Athlete Medical Station` …), or one of the names the project's TEAP prints, if it was imported first. Without that list, the project's TEAP names alone make the vocabulary. *Similar* means 80 % or better after normalisation, or a whole known name (of twelve characters or more) inside the label; a cable-length tag in front of the room (`L9m (31ft) Venue Media Centre (VMC)`) is stripped. Everything else a containment sheet prints — *CABLE GOES UNDER BRIDGE*, *L145m (476ft)*, *STRM SEWR*, the pathway legend, the drawing index, the survey abbreviations — is not a space. Left out by explicit rule, as before, are the title-block column (legend, client-group key, general notes, revision table), the status stamps, the scale bar, the sheet references and the NER markers themselves. A space read this way comes in **with no code**, with evidence naming the known name it resembles; in the matching step it can only match by name, so it always lands on the review queue. A sheet with codes is unchanged. On a sheet ticked as a **key plan** (below) no space is read at all.

If the import table shows `Spaces` at zero on a book with named rooms, the drawing carries no room labels as text (a scanned plan, for instance), and the matching step will refuse to run and say so.

KEY PLANS ONLY ILLUSTRATE

The overview sheets of a book — *KEY ENLARGED PLAN - LEVEL 01*, *KEY FLOOR PLAN* — repeat, at a small scale, the rooms of the detail sheets. The importer recognises them by the **DRAWING TITLE** in the title block (lower right): any title with `KEY` … `PLAN` is a key plan — and so are the other sheets that only illustrate: the *DRAWING INDEX*, the *VENUE INFORMATION & GAMES MAP*, the *ABBREVIATIONS*, the *TYPICAL … SOLUTIONS*, the symbols sheet, a cover or title sheet, general notes and legends. On a key plan: **no space is read** — nothing drawn, nothing in the search, nothing in the TEAP × drawing list — and it takes part in no crossing; no NER is placed on it; a NER written on it (a reviewer's red comment) comes in as a **reference** (`⋆NER★` / `⋆NER`), drawn with its circles and counted nowhere. The page stays in the viewer for references and illustrative areas. The ticks are reviewed on the Scale screen — one box per page, `Apply key-plan ticks` — and the `Key plan` box in the Drawing viewer does the same for one page.

NERS THE DRAWING ALREADY STATES

In the accommodation-matrix taxonomy (the AAM workbooks) `TEC.010` is a *Network Equipment Rack - Critical* and `TEC.020` a *Non-Critical* one; the older books filed them as `TEC.003.01` / `TEC.003.02`, and both spellings are read (`TEC.003-001`, the Mobile Equipment Room, is not a NER). The import **reads** what the book says about its NERs — the team's own annotations (*Critical NER 8* written on a sheet), those rooms, the *Network Equipment Rack* labels and the NER block symbols — and **places none of them**. That is a step of its own, `Place NERs from the drawing` on the Scale screen, to run once the key plans are ticked and the scales confirmed (the 12 m rule below needs the scale: on a page with none it is applied as 170 pt, which is 12 m only at 1:200). It creates the NERs nobody has drawn yet — for you to check and validate, instead of drawing them all from scratch — in this order: the team's annotations, then the coded rooms, then the labels, then the symbols:

- **What is read:** the code in any of its spellings (`TEC.010-003`, `TEC.020-003`, a bare `TEC.010`; `TEC.003.01-05`, `TEC.003.01.005`, `TEC.003.02` on older books) or, with no code, the label *Network Equipment Rack (NER) - Critical / Non-Critical* or, as the newer books print it, *Network Equipment / Rack - Non-Critical* (two lines, no bracket — a lone *Rack - Critical* with no *Network Equipment* over it is not a room). A sentence in a note or the legend is not a room.
- **The NER block symbol:** the newer books draw most racks not as rooms but as a small orange square with the word *NER* inside, in a dashed rectangle in the area it serves. The word is not text on those sheets (the CAD plots it as strokes), so the application recognises the block by its *shape*, against a reference block, whatever its size or angle; the colour decides nothing. A symbol is a NER **at the square's centre**, **non-critical with its criticality pending** (the symbol does not say); a symbol drawn inside a room already read is that room's rack, not a second one; a symbol at the same position as a symbol on another sheet (a ground-floor rack repeated on the floor above) is created and asks whether it is the same rack. Its evidence names the shape distance and the stroke count.
- **Where the NER goes:** when the label has an arrow to the room, the leader is followed to the arrow's tip and the NER goes there; on a label stacked with another room's name (*… Critical / Commentator Control Room (CCR)*) the arrow leaving the NER's lines counts. With no arrow, the NER sits on the label.
- **Criticality:** the code decides (*-Placeholder* with `.01` is critical); with no code, the label's words. If the drawing does not say, the NER comes in **non-critical with its criticality pending** — the `criticality not stated` tile on the NERs screen filters those rows, and the row can only be confirmed once you choose.
- **Where there is already a NER:** a NER written on the drawing or drawn by you within 12 m of the room, on the same sheet, **is that room** — no second NER; the code is recorded on it, and if its criticality disagrees with the drawing's the row asks which is right. One existing NER may answer for several rooms near it (racks labelled side by side in one cabin, one NER drawn for them): no new NER is created near one that exists.
- **The same code on two sheets** (two overlapping detail sheets) is one NER, with a marker on each sheet — when the code carries a room number (`TEC.020-003`). A bare class (`TEC.010`) printed on two sheets is two racks.

- **Name:** in sequence, by the application's own rule — `NERc_01.1`, `NERc_02.1` ... for a critical room, `NER_20.1`, `NER_21.1` ... for a non-critical one (the critical series starts at 01.1, the non-critical at 20.1; the `.1` is the rack — a second rack beside it is `.2`, renamed by hand) — in reading order within each group — the annotations, then the coded rooms, then the labels, then the symbols — by page, then top to bottom; what the sheet printed (`TEC.003.01-05`, *Critical NER 8*) is kept as *printed as*. Rename them on the NERs screen as you like.
- **Running it again** adds only what is still missing: every candidate finds the NER already there — a NER you **removed** included: its markers still hold the spot, and the room is not created again. A duplicate that slips through is yours to delete in the Drawing viewer (`Delete only this marker` when the room is drawn on other pages too). A NER whose drawing you take off a sheet - with `Delete only this marker` or by a move on the composite map - is **not drawn back on that sheet**, wherever the sheet prints its code; where the application knows only the sheet and not the spot, the whole sheet is kept off for that NER.

Every NER created this way comes in **not confirmed**, with the origin `from the drawing` (an annotation) or `from the room label` (a room or a symbol) and, in its evidence, the label as printed — *NER* for a symbol — and how its position was found. The notice at the end of the import says how many annotations and rooms were read; the notice at the end of the placing step, how many NERs were placed.

2 · TEAP (XLSX / XLSM / CSV)

It accepts `.xlsx`, `.xlsm` and `.csv`. An old `.xls` is not offered by the file picker and, if sent anyway, is refused with *an old .xls workbook cannot be read...*: open it in Excel and save it as `.xlsx`. Macros are never executed and formulas are read as their cached value. After the upload the application profiles every sheet and proposes the sheet, the header row and the mapping; three blocks appear:

Sheet and header row

CONTROL	WHAT IT DOES
`Sheet` SELECT	Each option reads `name — n rows (header row n)`. Changing the sheet resets the header row and the mapping to what was proposed for it and re-reads the venue list.
`Header row` NUMBER	Which row holds the column headings. Changing the number does not re-read the headings: the mapping lists keep those of the proposed row, so check that the chosen columns exist on the new row.
`Reload venue list` BUTTON	Re-reads the venue column with the current sheet, header row and mapping.

Venue

If the sheet holds more than one venue, a banner appears (*This sheet is a master allocation plan: n venues, n rows...*) and the import is **blocked** until you choose. Use the `Venue filter` select — each option reads `CODE · Name — n rows`; the project's code is suggested. Below it, *Will import n of n rows* confirms the cut. The `import every venue` box releases the block by importing all of them. The filter works on the **code**: map

`venue code` (with only `venue name` mapped, the select does not filter). With no venue column mapped, the whole sheet is imported. A **CSV** has no sheet and no venue filter: the header is always the first line and the whole file is imported, every venue in it — for a programme-wide TEAP, use the XLSX.

> THE APPLICATION'S MOST EXPENSIVE MISTAKE
>
> Ticking `import every venue` on a programme-wide TEAP pulls dozens of venues into a project that holds the drawing of *one*. Because the space-code taxonomy is shared across venues — `SPT.016.01` exists in dozens of places — the matching pairs other venues' rooms against your plan with high confidence, and **nothing in the matcher checks the venue**. Measured on a real project, of 2,873 matched rows only 74 (2.6%) belonged to the right venue; every coverage figure after that is noise that looks like a result. Always pick the venue.

Column mapping

One select per canonical field, each starting at `— not mapped —` or at the heading the application recognised. The fields: `venue code`, `venue name`, `building`, `level`, `zone`, `fa code`, `functional area`, `space code`, `space name`, `product code`, `product description`, `product category`, `quantity`, `network ports`, `wireless required`, `criticality`, `sport code`, `comments` and `serving ner` — the last one optional: the column in which a TEAP names the NER serving the space, when it has one. See the rule under Demand per space.

`space name` is mandatory — it is the key the drawing match is built on. Without it, `Import TEAP` stays disabled with a matching warning. The `Save this mapping as a reusable template (optional)` box stores the mapping under a name; saving again under the same name **replaces** that template. To reuse one on another venue, the `Apply a saved mapping template` select, just above the venue section — it appears only once a template has been saved; each option reads the template's name and, in brackets, the sheet it was saved from — puts the template's columns on the mapping — only those this sheet has; the others are named in the notice and left as they were — and its header row. When the import finishes, the notice says how many rows went into how many spaces and how many ports, how many earlier decisions were carried across (on a re-import), and the screen jumps to `Scale`.

On import, a space whose name, code or functional area mentions *photo*, *photographer*, *photo position* or *PHT* receives the `photographer` demand type (70 m limit); every other space receives `general`. The type can be changed later, row by row, on Demand per space. Criticality comes first from the pair of X-ticked columns `Critical NER` / `Non-Critical NER`, which the application finds on its own (they are not on the mapping list); failing a tick, from the `criticality` column, read as it is. A space ticked in both columns is reported as a contradiction in the document, and the application does not pick a side; a space with no criticality at all gets a warning asking for confirmation.

3 · Project rules

CONTROL	WHAT IT DOES
`General limit (m)`	The default distance limit. 90 m by default.
`Photographer limit (m)`	The limit for photographer areas. 70 m by default. It is the stricter of the two: it cannot be longer than the general limit.
`Near-limit threshold (0–1)`	The fraction of the limit at which a space starts being reported as *Near the limit*. 0.90 by default (81 m of 90).
`Routing slack factor (≥ 1)`	The multiplier between the straight line and the real cable. See the distance rules.
`Display units`	`metres`, `feet` or `both`. Affects how distances are displayed on screen; the limits are still set in metres and the exports come out in metres.
`Fibre runs to the TER — per critical NER` `Fibre runs to the TER — per non-critical NER`	How many **fibre runs from the TER** each NER needs: `2` per critical NER (A and B paths) and `1` per non-critical NER by default. A **count**, not a length — the application measures no fibre route. It feeds the `Fibre runs to TER` column of the *NERs and ports* sheet and the totals on the *Summary* sheet (Reports).
`Save rules` BUTTON	Saves the project's rules; the next analysis uses them. An out-of-range value (a blank or 0 limit, a photographer limit longer than the general one, a threshold outside 0–1, slack below 1, fibre runs above 16) is not saved, and the red *Rules not saved* notice says why.

A NEWER TEAP REPLACES THE OLDER ONE

A project has **one** TEAP. Importing another file — or the same one again — replaces every TEAP space of the project with those of the imported file, and **carries your decisions across**, found again by the space key (venue + space code; with no code, the space name): the room tied by hand, the confirmation, the `— none —`, the NER chosen under `Served by`, the NER-on-another-level declaration, a hand-set `photographer` type and the `Same space` / `Not the same` answers. A space that no longer exists in the new TEAP goes with its decision; a new space comes in undecided. After importing, run `Run matching` and `Run analysis` again. An import that finds **no** row (a venue filter that matches nothing, the wrong sheet) is refused with *nothing to import… The TEAP already in the project was left as it was*, and nothing is replaced.

Which quantity column. A TEAP with more than one quantity column (`TEAP v2.0`, `NEW QTY`, `Updated QTY`) is read from `Updated QTY` — which is what the workbook's own README says to use; the other two only feed it. Check the `quantity` field in the mapping before importing. Ports always come from the `Network Ports` column when it exists.

STEP 2

Scale

Detected scales are proposals. Definitive results stay blocked until you confirm each page.

COLUMN	SHOWS
Page	Page number; below it an amber `OCR needed` pill when the page is scanned.
Key plan CHECKBOX	Ticked, the page **only illustrates**: no space is read off it, it takes part in no TEAP × drawing crossing and no NER is placed on it (see Key plans only illustrate). It comes ticked from the title block — `KEY … PLAN`, the drawing index, the venue information, the abbreviations, the typical solutions — and the row reads `KEY`. Tick or untick as many pages as needed; a `*` marks a change not applied yet, and the `Apply key-plan ticks (n)` button above the table applies them all in one act (`Undo ticks` drops them): a page ticked loses its spaces and any TEAP tie to them (recorded in the audit), a page unticked gets its spaces read off the book, and the matching runs once at the end. The notice says how many spaces went and came.
Sheet / level	The drawing number read off the title block (`VFF-401-CSL-O`), the sheet title and, when the title states one, the level (`LEVEL 01`).
Detected	The scales the application read off the sheet.
Applied	The scale in force, as a label (`1:200`, `1/8" = 1'-0"`). On import the first detected scale is applied: an architectural one first, then a ratio next to *SCALE*, then a bare ratio.
Source	The kind of notation (`ratio` or `architectural`) — it does not say whether the scale was detected or typed.
Status	A `confirmed` (green) or `not confirmed` (amber) pill.
Set (ratio or text)	A free box. A bare number is the denominator (`200` = 1:200); any other text is parsed as a scale expression (`1/8" = 1'-0"`).
Confirm BUTTON	Applies whatever is in the box *and* confirms the page. With the box empty it uses the scale in the `Applied` column, rewritten as a ratio (`1/8" = 1'-0"` becomes `1:96`) — and if there is none, it refuses with *"Nothing to confirm on this page yet."* There is no "unconfirm": a page confirmed with the wrong value is corrected by typing the right one in the box and confirming again.

Above the table, `Confirm all pages (n)` confirms **every page that already holds a scale** — detected or typed — in one click, after asking. Pages with **no scale at all** are left alone and named in the notice: the only honest thing to do with them is to keep blocking their results until somebody types the ratio in the row. Beside it, *n of n confirmed* and, in amber, how many pages still have no scale.

> NO READABLE SCALE ON THE SHEET
>
> Read the scale off the title block of the PDF itself, type the ratio (`200`) or the text (`1/8" = 1'-0"`) in the row's box and confirm. Until a page has a scale, every space on it carries the status *Scale not confirmed* and no definitive distance is computed.

NERs from the drawing

The second card of the screen, and the third step of the flow: **key plans → scales → NERs**. `Place NERs from the drawing` creates the NERs the drawings state (see NERs the drawing already states): the team's annotations first, then the coded rooms (`TEC.010` / `TEC.020`, or `TEC.003.01` / `.02`), then the *Network Equipment Rack* labels with their arrows, then the NER block symbols — never on a key plan, never within 12 m of a NER already on the sheet. "Where a NER is" means both where its marker is now and where it

was first put: a NER read off a room and then dragged onto the real rack still answers for that room, however far it was dragged, and the room is not created a second time. It asks first, and names in the question what is still open: key-plan ticks not applied, detail pages with no confirmed scale (the 12 m rule needs it — without a scale the same-place test is applied as 170 pt, 12 m at 1:200, which is wrong on any other sheet; a detected scale not yet confirmed is used as it is). It runs as a job that walks every detail sheet's strokes (the symbols have no text to announce them — a second or two per sheet, the progress shown page by page); the notice says how many NERs were placed, how many came from the annotations, how many rooms were read and how many already had a NER beside them. The line beside the button keeps the count of key plans, of detail pages with no confirmed scale, and when the step last ran. Running it again only adds what is still missing. A second press while it is still running is refused (*already running for this project — wait for it to finish*), so a double click never places a NER twice.

STEP 3

Drawing viewer

The real plan, with markers, spaces, routes and the coverage overlay — and the ability to edit all of it directly on the drawing.

The top strip and the toolbox

The strip across the top holds only what is about *where you are*: the page, whether it is a key plan, and the search. Every tool, and the view switches, are in the **toolbox** — a column down the left of the sheet, in groups: `select` ; the NERs; the rooms that only show; the Wi-Fi point; the illustrative tools (violet); routes and shafts; the ruler (`measure`); the eraser; and at the bottom `Fit` , `Radii` , `Spaces` , `Register NERs` , `Run analysis` and, under it, when the analysis shown was run. The `‹` in its corner folds it to a `Tools ›` tab; the choice is remembered in this browser. The viewer **comes back to the page you left**: leaving for another screen and returning lands on the same page, per project, in this browser.

TOOL	WHAT IT DOES
◀ · Page n - drawing number · ▶ SELECT	Chooses the page — the arrows step to the previous and the next page, the select picks one, and `8/47` says where you are. The select reads `Page 1 - VFF-401-CSL-O · LEVEL 01` — with several series in the project, `400 · Page 6 - VAA-411B3-VBV-O · LEVEL 01` , the series first and the page within it, every series one after the other: the page number the application uses, the **drawing number** the design team uses (read from the title block's *DRAWING NUMBER* field) and the level when the *DRAWING TITLE* states one; `· KEY` on a key plan. A sheet with no title block shows just `Page n` .
Key plan CHECKBOX	Says whether the page is a **key plan** — illustrative only (see Import). It comes ticked when the title block's *DRAWING TITLE* says `KEY` … `PLAN` ; it turns amber when on, and the page selector shows `· KEY` . It is there to correct a title block read wrong: ticking or unticking asks for confirmation; ticking releases every tie between the TEAP and a room on this page (including those you confirmed, recorded in the audit) and runs the matching again; unticking brings the page's rooms back into the crossing and runs it again too. NERs and Wi-Fi points already drawn on the page stay — the notice says which still count, to delete or replace with `*NER` . On a key plan, `+NER★` , `+NER` and `+Wi-Fi` are disabled and `Spaces` is off — a key plan has no spaces. The same ticks, for every page at once, are on the Scale screen.

TOOL	WHAT IT DOES
search a name… SEARCH BOX	Finds a name on **any page**: a room read from the drawing (by its code, its name or the TEAP code tied to it), a NER or any other marker (wireless points and references included), a named area. Every word typed must appear, in any order, ignoring case and accents. As you type the list appears (up to 15), with the type and the page; exact name or code first, then those that start with the text. `Enter` goes to the first in the list, a click to the one chosen: changes the page, centres, zooms in, selects it in the Inspector and rings it in orange for a few seconds. The `Drawing` button on Exceptions opens the viewer already searching for the room.
select	Navigation mode. Only in this mode can you drag the plan and move unlocked markers.
+NER★	The next click creates a critical NER. Asks for the label, offering the next name of the critical series (`NERc_03.1` when `NERc_01.1` and `NERc_02.1` exist); anything typed over it stands.
+NER	The next click creates a non-critical NER, offering the next name of the non-critical series (`NER_20.1` , `NER_21.1` …).
A name already used	When the label typed for `+NER★` or `+NER` is the name of a NER that already exists (ignoring case, accents and any spaces or punctuation: `NER 21` = `ner21` = `NER-21` , but `NERc 01` ≠ `NER 01` and `NER 3` ≠ `NER 03`), the app asks, saying which pages it is on. **OK**: it is the same NER, drawn on this page too — the marker becomes one more occurrence of it, with its name and classification, and it stays **one** row on NERs, counted once. **Cancel**: it is a different room — the app asks for another name, and asks again if that one exists too. A removed NER is left out of the question (restore it on NERs if it is that one), and renaming a NER to a name that already exists asks nothing. A wrong **OK** is undone with `Separate` , in the row's detail on NERs.
+TER +IDF +COW +DIVA	The next click creates a TER, an IDF, a COW or a DIVA node. **Visual only**: they appear on the plan and on the exported PDF, can be moved, renamed and deleted, and never serve a space or get a reach circle. The DIVA node is also read off the VDB on import (*DIVA Node*, *DIVA NODE*, *DIVA Node Rack*).
The illustrative group (violet buttons) — everything these buttons draw appears on the plan and on the marked-up PDF and **counts nowhere**: not on the NERs list, no TEAP link, no port, no figure changed. The different colour is so that `*NER` is never taken for `+NER` .	
*NER★ *NER	Each click places a **reference to a NER** on a key plan (the overview sheet, *key enlarged plan*, that shows the whole venue at a small scale): a dot in the NER's colour, critical or not, with a **dashed ring**, the name of a NER from the NERs list (chosen in the select that appears in the toolbox while the tool is armed, or typed) and the **90 m and 70 m circles to this sheet's scale**. It is a pointer to a room drawn on another sheet, and nothing else: not on the NERs list, no TEAP link, no port, counted nowhere — like a TER or a COW, but with the circles, because that is what it is drawn for. The tool stays armed; choose `select` to stop. In the Inspector, `Points at (NER of the register)` renames it after another listed NER, `Kind` switches among the five reference kinds (critical NER, non-critical NER, TER, COW, DIVA node). It comes out on the exported PDF with its circles and has a row of its own in the legend.
*TER *COW *DIVA	The same for a **TER, a COW or a DIVA node**: each click places a reference on a key plan, with a dashed ring in the room's colour and the name you type (`TER` , `COW` , `DIVA Node` by default). **No circles** — the real room has none either. The tool stays armed. In the Inspector, `Kind` switches among the five reference kinds.
+area	Draws a **named area** to point something out on the sheet — the BTOC, the TER room, a compound. Click each corner (the toolbox counts them); **double-click the last corner or press** `Enter` to close the outline and be asked for the name; `Esc` drops what was being drawn. At least three corners and a name are needed. The area shows as a faint slate-blue fill with an outline and the name on a tag at its centre, and comes out the same on the marked-up PDF, with a row of its own in the legend. It is **illustrative only**: nothing is measured or counted from it. Click the area to open it in the Inspector (rename, annotate, delete); it also goes with `Delete` or the `delete` tool. The tool stays armed for the next area.

TOOL	WHAT IT DOES
+Wi-Fi	Each click places a **Wi-Fi point** — the Wi-Fi sign in light blue, numbered `WIFI-1`, `WIFI-2` … (one kind: indoor and outdoor are not told apart). It marks a space that has wireless; it is **not an access-point count** — the number of access points comes from the wireless design. The tool **stays armed** for the next point; choose `select` to stop. A Wi-Fi point is not a room and **takes no port**: it marks the TEAP space it sits in, whose wireless port is that space's `DWISA` line (see Demand per space). At the next analysis it is measured to the NER of its space — or, tied to no space, the nearest NER within the general limit — so the 90 m rule is checked where the point is drawn. On placing, the app proposes that space: the nearest room on the same sheet that is matched to the TEAP, up to the general limit; the notice says which. Drag it to move it (the proposal follows). The points `Place Wi-Fi from the TEAP` puts down on the Wi-Fi step are the same thing, already tied to the space they were placed for — the Inspector reads *placed from the TEAP* — and a move does not change that tie. In the Inspector you rename it, choose or clear the space under `TEAP space` and delete it; it also shows the NER serving it, the distance and the status, from the last analysis. It also goes with the `Delete` key or the `delete` tool.
route	Traces a **cable route** — the path the cable will follow, along a corridor's tray, for example — **that the analysis uses**. Click where the path starts, then each corner along it: a dashed line follows the pointer with the length so far and the length of the leg being drawn. **Double-click the last corner or press `Enter`** to end it; `Backspace` takes back the last corner; `Esc` drops the whole route. Clicks land on the sheet even over a marker, so a route can start on a NER. The route is saved as one, named `R1`, `R2` … in the project, and shows in teal with its name and length at its end (`R1 · 56.4 m`), on this sheet's scale. A corner placed close to another route **snaps onto it** — on its corner, or anywhere along it — and the two routes are then joined. How the analysis uses it: a space and a NER that are **each within about 15 m** of the traced routes of their sheet are measured along them — from the space to the nearest point of a route, along the routes, to the nearest point to the NER — instead of the straight line; the two short straight pieces at the ends are added. The space's assumptions in the Inspector say how much was traced and how much is the straight lead-in (*24.7 m along the traced route + 3.5 m straight lead-in at the two ends*), and the method reads `route`. A space or a NER further than that from every route is measured in a straight line as before. Click a route to see it in the Inspector (name, segments, length; the whole route turns orange) and delete it there or with the `Delete` key; it goes whole. The new distances appear at the next `Run analysis`. The tool stays armed for the next route.
vertical	One click, then the sheet the shaft connects to and the vertical length in metres. With one drawing book the sheet is its page number (`12`); with several books it is the drawing number and the page (`600 12`), or the drawing number alone for its first page — a page number alone would not say which book, and the tool says so instead of guessing. This is how cross-floor service becomes measurable — see Demand per space. The shaft shows on both pages it joins, as a teal square with the page it leads to and its length beside it (`shaft ↑ p.12 · 4.5 m`); click it to see it in the Inspector and delete it.
measure	**The ruler** — a quick measure on the sheet, for your own analysis. Click the first point, then each corner: a dashed magenta line follows the pointer with the length so far. **Double-click the last point or press `Enter`** to end it; `Backspace` takes back a point; `Esc` drops it. The length is read on **this sheet's scale** and shown in metres, feet or both, as the project shows lengths; on a page whose scale is not confirmed it says so beside the figure, and on a page with no scale the tool refuses. Clicks land on the sheet even over a marker, so you can measure from a NER. Finished measures stay on their sheet while you work, and **nothing is saved**: they are not routes, not used by the analysis, not in any export or report, and they are gone when you leave the viewer. Take one away with the `delete` tool, or all of this sheet's with `Clear measures (n)` in the toolbox.

TOOL	WHAT IT DOES
`delete`	**The eraser.** Armed (the button turns red), every click on a marker — NER, TER, IDF, COW, DIVA node, reference —, a wireless point, a named area, a route, a shaft or a measure **deletes** what was clicked. It stays armed until you choose `select`. A wireless point, a measure and a route of one segment go without a question, because redrawing them costs a click or two — a Wi-Fi point tied to a TEAP space then stays deleted: `Place Wi-Fi from the TEAP` does not put it back, and the notice says so; a route of several segments, a marker, an area and a shaft ask first; a route goes whole. A locked marker refuses: unlock it first. Deleting a NER's marker does the same as the Inspector's `Delete`: **the NER leaves every page, the NERs list, the analysis and the export** — the question names the other sheets it is drawn on; the row is kept under `Removed (n)`, with the page it was deleted from as the reason, and `Restore` brings every page back.
`Fit`	Resets the view.
`Radii`	Toggles the reach circles. Only drawn when the page has a scale.
`Spaces`	Toggles the spaces and their labels.
`Register NERs (n)`	Toggles the rooms on the NERs list that have a position on this page but **no marker on the sheet**. The number is how many there are. Each shows with its name, *not on the sheet* and only the blue general-limit circle. The button **only appears when there is one**: since a NER you draw is the list row itself, that is rare, and a button with no effect would mislead.
`Run analysis`	The **same analysis** as `Run analysis` on Demand per space: it runs in the background and is saved as the project's analysis, which every screen and export reads. The viewer **always shows the last saved analysis**, from the moment it opens — run here or on Demand per space: every space matched to the TEAP in the colour of its status and a dotted line to the NER serving it, on the same page; with a NER selected, the Inspector lists the spaces it serves, the ports assigned and the utilisation; a space or a Wi-Fi point shows its NER, distance and status. Under the button, *Last run* says when that analysis was run; when something changed after it — a NER moved, a route traced — it turns amber and reads *changed since the last run (n)*, and the colours are those of the last run until the button is pressed again. Every row of the analysis carries the drawing room it was measured from, so the paint covers every match — exact, by similarity or by hand. A space still grey is a drawing space no TEAP line is tied to, or one the last run did not reach.

The wheel zooms about the pointer; dragging pans (only in `select`). Space labels only appear above 0.5× zoom. An amber `scale not confirmed` pill stays visible whenever the page is still unconfirmed. If the NERs list holds rooms with **no position at all**, the top strip says in amber how many (*n NER(s) in the register have no position on this drawing set. Place them with the +NER tools*) — never an invented dot in the corner of the sheet.

ERASING WHAT YOU DREW

Everything this screen puts on the sheet comes off through this screen, three ways: **select it and press** `Delete` (or `Backspace`) on the keyboard; select it and click `Delete` in the Inspector; or arm the `delete` tool and click what should go. It covers markers of every kind, wireless points, named areas, routes, shafts and measures. The key is ignored while you type in a box such as the `Name` field. Every removal goes to the audit log, and the numbers only change at the next `Run analysis` (here or on Demand per space). Deleting a wireless point, a route, a shaft or a named area brings a notice to re-run the analysis (*Re-run the analysis to see the effect*); deleting a TER, an IDF, a COW, a DIVA node or a reference only says it left the drawing.

Deleting a NER deletes the NER, not one drawing of it. A room is often drawn on several sheets — a level plan and two details — and the register holds it as one NER with those pages. `Delete` on any of its markers removes the room from **all** of them, from the NERs list, from the analysis and from the export, in one act (the saved analysis is out of date until the next run — *Demand per space* says *changed since the last run*, and an export made before that run runs the analysis again first), exactly as `Delete` on the NERs screen does; the question says *Delete the NER …? It is also drawn on p. 12, p. 15*. The row and its markers are kept, hidden, under `Removed (n)` on the NERs screen, where `Restore` puts it back on every page. A removed NER also **keeps its place on the sheet**, and its code on every sheet and book: `Place NERs from the drawing` will not create a room there again, nor from the same code printed elsewhere, nor from the NER block symbol drawn in the same room — a room removed before its markers were kept (with no code to hold its place) included, by the position the NER list keeps for it. For the one case where only this drawing is wrong — the same room placed twice on one sheet — the Inspector offers `Delete only this marker` when the NER is on more than one page: that marker goes, the NER stays on its other pages (and is retired only if it was the last one). The spot that marker held stays the NER's: `Place NERs from the drawing` takes the room printed there as this NER — it creates no other NER there and does not draw this one back.

A wireless point can also be deleted on the Wi-Fi step, with the `Delete` button of the *Wireless points* table; a NER, also on the NERs screen, where `Removed (n)` → `Restore` brings it back.

What the plan draws

- **Markers** — a filled circle in the kind's colour (see the notation), with a hairline black ring and the name beside it, bold on a **translucent white tag** so it stands out from the CAD linework underneath. An unconfirmed marker gets a dashed amber ring; a reference (`⋆NER`, `⋆TER` …), a dashed black one and a name two points larger than a room's, to be read from across a key plan (the marked-up PDF does the same).
- **Named areas** (`+area`) — a slate-blue polygon, faint fill, the name on a tag at its centre. Illustrative: nothing is measured from them.
- **Listed rooms with no marker on this sheet** — drawn like a marker, in the colour of their criticality, with *not on the sheet* under the name and only the blue general-limit circle, while `Register NERs` is on. That is a fact about the drawing, not a different kind of room.

- **Spaces** — dots with the code or name, in the colour of their status in the last analysis; grey when the last analysis did not reach them.
- **Routes** traced in teal, each with its name and length at its end; **measures** in magenta, with their length; **reach circles** in blue (general limit) and purple (photographers), both **divided by the slack factor**, around every NER and every **NER reference** (`*NER`); a listed room with no marker on this sheet gets only the blue one. A TER, an IDF, a COW and a DIVA node — and their references — get no circle: a radius around one would promise a coverage that is not there.
- **The legend**, in the bottom-right corner: the six room kinds, the Wi-Fi sign (*Wi-Fi — space with wireless*), the reference on a key plan (a dot with a dashed ring), the named area, the orange *Match not confirmed* rectangle, the limits, the slack, the page's scale and the disclaimer. `hide ▾` at its top folds it to a small `LEGEND ▴` tab, which opens it again; the browser remembers the choice. The exported PDF's legend is a separate copy of this table, kept by hand in the exporter.

> MOVING A ROOM ON THE PLAN
>
> In `select` , drag the marker. It works for a NER off the sheet and for a room the sheet does not carry. The move is audited (`ner.moved` , from where and to where); the marker's origin **does not change** — a marker read off the drawing and moved by hand is still `from the drawing` . A **locked** room refuses to move — that is what the lock is for; a drag on it falls through to the plan, which pans as if you had dragged the background (`Fit` recentres). Moving an unlocked marker leaves the sheet where it was.

Inspector

The right-hand panel. With nothing selected it explains the tools. Click a **marker**: the `Name` box (rename — for a NER the name on the NERs screen follows; the name of another NER asks whether they are the same room, as on the NERs screen), the `Kind` select (critical NER, non-critical NER, TER, IDF, COW, DIVA node; on a reference, the five reference kinds), page, origin (*from the drawing* / *drawn by you*), confidence, whether it is confirmed, the detection evidence, a NER's capacity (total, used and reserved ports), and the `Confirm` / `Unconfirm` , `Lock` / `Unlock` and `Delete` buttons (on a NER drawn on more than one page, also the line *This NER is drawn on n pages* with `Delete only this marker`). From the last analysis it also shows the served spaces and the utilisation.

On a **NER** the Inspector also has `Indoor / outdoor` — where the rack stands, the same field as on the NERs screen. A **Wi-Fi point** has one too, under its `TEAP space` : `Indoor` (a covered area) or `Outdoor` (an open area), the same field as in the *Wireless points* table of the Wi-Fi step. On any marker that came in with a printed label, the line `Printed on the drawing as` shows what the sheet carried — for a NER, the label it had before it got its sequential name.

Changing the `Kind` of a NER's marker to TER (or IDF, COW, DIVA) takes that marker off the NER. If it was the room's last marker, the room leaves the NERs list — the row is retired with the reason and can be restored; a NER drawn on other pages too keeps its row. Switching between critical and non-critical NER changes the whole room, every marker of it, and confirms it — as the `Criticality` select on NERs does. Changing a TER to NER puts it on the list at once. A TER, an IDF, a COW and a DIVA node are managed **here, and**

only here: renaming, re-classifying, confirming, locking and deleting happen in the inspector, with the plan in front of you.

Click a **space**: name, code, page and — from the last analysis — the serving NER, the distance against the limit, the method, the ports, the status and the list of assumptions and warnings. Click a **listed room with no marker**: name, code, page, criticality, which space it sits in, why it is there and the outstanding validations.

On a **NER read off the room label** (see Import), the line under the name reads `from the room label` , `Printed on the drawing as` shows the label as printed, and in amber come the questions it still carries: *criticality:* (the drawing does not say, or says otherwise than the NER that was already there) and *position:* (no arrow could be followed — the NER sits on the label). Dragging the NER onto the room answers the position question; choosing the criticality (here under `Kind` or on the NERs screen) answers the criticality one.

Click a **named area**: the name (`Name` , editable), `Notes` , the page, how many corners it has and `Delete` .

The panel has two tabs: `Inspector` , described above, and `Naming` — the naming panel, to read while renaming (see The naming reference); it decides nothing. The panel takes 316 px from a drawing that needs every pixel. The `›` button in its corner collapses it; collapsed, it becomes a `‹ Inspector` tab on the edge. The choice is remembered in this browser.

STEP 4

NERs

One screen, one list: one row per NER, inherited from the drawing or drawn by you. Everything decided about a NER is decided here.

> ONLY A NER SERVES
>
> A **NER** is the room this study is about: editable, classifiable, the one that serves the TEAP's spaces, and the only kind that appears in the `Served by` list. **A TER, an IDF, a COW and a DIVA node appear on the drawing and stop there.** They are read off the VDB, drawn, moved, renamed and deleted in the Drawing viewer. They are not rows on this list, never serve a space, never enter the analysis or the workbook's *NERs and ports* sheet, and get no reach circle. The line under the buttons (*Also on the drawing, managed in the Drawing viewer*) says how many there are — TERs, IDFs, COWs, DIVA nodes and the references drawn on key plans — and on which pages.
>
> The refusal is the server's, not only the dropdown's: the API answers *“only a NER can serve a space”*.

The tiles and the buttons at the top

CONTROL	WHAT IT COUNTS OR DOES
NERs	Live rooms on the list — excluding removed and consolidated ones.
not confirmed	How many still lack the `Confirm` tick. Amber while above zero.
criticality not stated	NERs whose label did not say whether the room is critical — choose in the row's select. **Clicking the tile filters the list** to those rows only (amber outline); clicking again shows them all.
capacity unknown	How many have no ports recorded. Blank is *unknown*, never zero.
Confirm all (n) / Unconfirm all	Confirms every NER not yet confirmed — the criticality and every marker of each. Asks first; a row whose criticality is *not stated* is skipped and named, never forced. One audit entry per room, exactly as the single click writes. `Unconfirm all` takes it back.
Lock all (n) / Unlock all (n)	Locks (or unlocks) every NER against being moved on the drawing.
only rows not confirmed yet	The one filter: shows only what is still to be signed off.
Possible duplicates (n) · Consolidation history · Removed (n)	Open, below the list, the duplicate queue, the consolidation history and the removed NERs (with `Restore`). The last one appears only when there are any.
Naming reference	Opens, beside the list, the naming panel to read while renaming — see The naming reference. It decides nothing. The `›` in its corner folds it; the choice is remembered in this browser.

The columns

COLUMN	SHOWS OR CONTROLS
Page	The page the room is on (`p. 7`) and the others it also appears on (*also on*); *no position* for a room with no position yet. In a project with several series each sheet is written as series and page in that series: `p. 400·5`, *also on* `400·12, 600·4`. **Every page number is a button**: it opens the Drawing viewer on that sheet, zoomed in on this NER and ringed in orange for a few seconds. The pages under `Markers on the drawing` (in `Details`) do the same for each marker.
Name	**One name.** Every NER the application creates — from the team's annotations, from a coded room (`TEC.010` / `TEC.020` / `TEC.003`), from a room label, from a NER block symbol — is named **in sequence**, by the application's own rule: `NERc_01.1` , `NERc_02.1` … for a critical room, `NER_20.1` , `NER_21.1` … for a non-critical one (the non-critical series starts at 20; the `.1` is the rack), in reading order; what the sheet printed (*Critical NER 8*, `TEC.003.01-05`) is kept under `Details` as *Label printed on the drawing* and as another name of the room. A NER you draw keeps the name you typed. Editable; saves on blur. Renaming here relabels the marker on the sheet (and on the exported PDF), and `was …` appears below. Renaming in the viewer's inspector does the same thing — the two never disagree. A locked room cannot be renamed. The name you give a NER is the name: nothing suggests, corrects or numbers it for you. Two NERs cannot share a name (case, spaces and punctuation aside — `NER23` and `ner 23` are the same): typing the name of another NER asks *… is already the name of another NER: … Is it the same NER? OK makes the two one NER, with every marker of both. Cancel keeps the current name — choose another one.* — `OK` consolidates this NER into the other, as Possible duplicates does, and `Cancel` puts the old name back. Two NERs already sharing a name carry the outstanding validation *name: another NER has the same name* until one is renamed or they are consolidated.

COLUMN	SHOWS OR CONTROLS
Space code TEXT	The room's **space code** as the AAM or the VDB writes it (`TEC.010-001`). It arrives filled with the code the drawing prints beside the rack, when it prints one; type to correct it or fill it in. Saves on leaving the box. It is column C (`Space code`) of the *NERs and ports* sheet of the workbook, next to the `Full name` .
Origin	`from the drawing` (a marker read off the VDB — the NER label written on the drawing), `from the room label` (created by the placing step from the NER room the drawing prints — a `TEC.010` / `TEC.020` / `TEC.003` code, the *Network Equipment Rack* label or the NER block symbol; in amber, to check) or `drawn by you` (placed by hand in the viewer). Click the column heading to sort by origin, which groups the rows. Otherwise the three are treated alike.
Criticality	A `Critical` / `Non-critical` select, always editable. It starts at what the drawing says (a label *Critical NER 1* arrives as critical, with the evidence under `Details`); `Not stated on the drawing` appears only for a marker whose label said nothing, and disappears once you choose. **Choosing is confirming**: the value goes to the coverage engine and the marker on the sheet is repainted in that colour.
Indoor / outdoor	Where the rack stands — `Indoor` , `Outdoor` or *not said*. Said by you, never read off the drawing; it decides the kind of rack the venue is sent, so it is the `Indoor / outdoor` column of the *NERs and ports* sheet of the workbook (with three *Totals* lines on the *Summary*). The same select is in the viewer's Inspector.
Capacity (ports)	Installed ports. Blank is *unknown*, never zero — and the app cannot fill it in: knowing which spaces a room serves gives the *demand*; installed ports are the *supply*, and only someone who knows the switch and patch build knows that.
Lock	Locks the room against being moved on the drawing. Locked, `Delete` and the name are disabled.
Confirm	A person signed off: the criticality is right and every marker of this room on the drawing is real, at its position. Unticking takes the signature back. An unconfirmed marker shows in the viewer with the amber ring.
Details (n)	Opens the row's detail; `n` is how many validations are still outstanding.
Delete	Asks for the reason and removes the NER from **every page** of the drawing, the analysis and the export. **Nothing is deleted**: the row, its markers and the reason are kept, and `Removed (n)` → `Restore` brings everything back — unless a live NER now carries the same name: then `Restore` says so and does nothing until that one is renamed. If other rows carry the **same name** — they are separate NERs: two answers of “not the same NER” when a marker was placed, or two rows renamed alike — a second question lists their codes and pages and asks whether to remove them too (*OK*) or only this row (*Cancel*); a row left behind stays on its pages. The same act from the Drawing viewer: `Delete` on any marker of the NER.

Each row's detail

On the left: `Outstanding validations` (what is still unanswered), `Identity and history` (internal id and code, which never change; the label printed on the drawing, shown once the name is no longer that; other names; the rename history with date, actor and reason; where the criticality was read from) and free-text `Notes` . What you write under `Notes` goes to the `Notes` column of the *NERs and ports* sheet of the workbook.

On the right, `Markers on the drawing` : one marker per row, with page, label, kind, origin, confidence, the detection **evidence** (the text read, the colour, the layer) and whether it is confirmed. `Separate` splits a marker into a NER of its own when a consolidation was wrong for that page — or when the **OK** in the Drawing viewer's same-name question was a mistake, or when the same code placed one NER in two different rooms. It asks for the **name of the new NER** (empty gives the next free name in the sequence); the group's own name and the name of any other NER are refused. The marker leaves the group for good:

the new NER is a row of its own, the two are recorded as **different rooms** (the duplicate list never offers to merge them again), and if the separated marker held the group's registered position, that position moves to one of the group's own markers — nothing of the old NER stays drawn where the separated marker is. The room code printed on the drawing stays with the group, so a later `Place NERs from the drawing` does not give it, or the other sheets that print it, to the new NER. Check the new NER's criticality on its row. It only appears when the NER has more than one marker.

> COLOUR DOES NOT CLASSIFY
>
> Classification comes from several signals, with the label text weighted highest. On real design books criticality is written in the label, **not encoded in the colour** — so anything the app could not read from the text is deliberately left unconfirmed, and your confirmation is what closes the question.

The naming reference

The application names every NER it creates by its own rule: a prefix, a two-digit number and a rack suffix — `NERc_01.1`, `NERc_02.1` … for a critical rack, `NER_20.1`, `NER_21.1` … for a non-critical one; two racks side by side are `.1` and `.2` of the same number. The **full name** in the workbook puts the project's venue code in front: `VVV_NERc_02.1`. **The name you give a NER is the name**: rename the room in the `Name` field as your project's naming document says, one at a time. The rename goes to the room's history as any other.

The `Naming reference` panel on this screen and the `Naming` tab of the viewer's Inspector are there to read while renaming, **and the application reads nothing from them**: they never decide a name. When the copy carries a naming table in its private reference folder (`backend\app\reference_private\naming_reference.json`), the panel shows that table; otherwise it states the application's own rule above and that the name you give a NER is the name. A team can add its own naming document there; it is shown for reference only.

Possible duplicates

When the same name appears on different pages — `NER 5` on 7 and on 8 — the app does **not** merge on its own: it scores the pair below the automatic threshold and brings it to this panel with the evidence that they are the same room (*“identical normalised label”*, *“same marker kind”*, *“appears on pages 7 and 8”*) and the differences found (*“level not stated on at least one page”*), side by side. The pill gives the score and whether the evidence is strong (*strong evidence*) or calls for judgement (*needs judgement*); below it the impact: how many markers change row and how many served areas are reassigned.

- `Same NER — consolidate` : the two become one NER with two markers, one per page, and the whole coverage analysis counts one room instead of two. Write the `Justification` (*what makes you sure*) first. Reversible with `Undo` under *Consolidation history*.
- `Different NERs` : the pair **stops asking**.
- Or leave it for later — nothing changes.

An identical label is **necessary but not sufficient** — venues reuse numbering series. It only merges by itself when the label and the marker kind are the same and the **level** is stated and the same on both pages, with nothing in conflict — the building need not be stated; a level or building conflict disqualifies; `NER 4` and `NER 5` are two rooms by definition; an **unknown** level is not agreement — it becomes a question, not an assumption. “In the same area” is something *you* know and the drawing does not state; that is why the decision is yours.

Consolidation history

Every link the app made on its own is here with its evidence, alongside every human decision: when, the decision (*merged*, *separate*), by whom and by which method, the rooms, the score, the evidence and the justification. `Undo` appears only on merges not yet undone, and returns the list exactly to its previous state.

Renumbering a NER

The usual case: remove NER 007 and rename the last room to “NER 007”. Both keep their internal identifier, and the history records the swap. A NER removed with `Delete` **leaves the comparison**: it never comes back by itself, raises no question and does not appear under *Possible duplicates*. A new marker with its label quietly becomes a new NER; the NER you removed stays under `Removed (n)`, where `Restore` brings it back. Its markers keep their place on the sheets: `Place NERs from the drawing` treats the spot as taken and creates nothing there.

STEP 5

TEAP × drawing

Match each TEAP row to a drawing space. Nothing low-confidence is confirmed silently.

> KEY PLANS ARE LEFT OUT
>
> A room on a **key plan** (see Import) is not offered to the automatic matching, does not appear in the `Drawing space` list and is refused if sent by hand (*this room is on a key plan…*). A key plan repeats the rooms of the detail sheets, and tying the TEAP there led to the right room on the wrong page. A tie to a room that comes to be on a key plan is released — even one you confirmed —, recorded in the audit (`matching.key_plan_released`, with the room and the page) and the matching looks for the room on a detail sheet; if it finds none, the line goes to `no room found`. The *same space on other pages* column ignores the key plans too.

THE TEAP RULES

The TEAP says whether an area is served. The check runs **from the TEAP to the drawing, never the other way round**: a room that is on the drawing and that no TEAP line asks for is not listed, not counted and not a finding — however likely it looks that it will need service one day. And a TEAP space whose lines add up to **zero network ports** asks for nothing a NER can give: it is matched silently and left off this queue, its tiles, Demand per space, Exceptions, the analysis and the workbook's main sheet. It is a filter, not a deletion — the scope is worked out from the imported lines, so a later TEAP that gives the space a port brings it back by itself.

One exception. A space with zero ports whose lines request *wired connections* (`Wired Connection …` , `RJ45`) with a quantity above zero **stays on every list**, at zero ports, with the reason written on the row. The TEAP counts the port on the device, not on the service; where users bring their own equipment there is no device line and the port is never counted — the photographer positions on the FOP and the press tribune are the cases. The application **invents no port** for those lines: it shows the space, measures the distance and leaves the count to be validated with the TEAP owner.

A BOOK WITH NO SPACE CODE

On a book that labels rooms by name alone (see How a space is recognised), the code steps do not exist: everything matches by **exact name, normalised name or similarity**. An identical name on a TEAP line with no code, answered by one room only, is confirmed by the matcher like an exact code; everything else reaches this queue for confirmation. Expect three things. *(1)* Several TEAP rows with one name (`SEC.005-01` to `-07` , all *Pedestrian Screening - Accredited (PSA)*) point at the same drawing label — pick the right room for each. *(2)* One name printed twice on the sheet (two concessions, two *Announce Position*) is legitimate ambiguity, not an error. *(3)* The rooms with the largest demand may not be labelled at all: a VDB that draws a pavilion as one block (*North Pavilion: Three Levels*) shows neither the press tribune nor the photo work room inside it. For those, the worklist lets you pick any room on the drawing, or the row stays unmatched until the book details the interior. Normalisation already treats *Centre* and *Center* as one word.

The order of attempts

```
exact code (number by number) → the code family (deepest first, then name, then free room)
```

The code comes first because the drawings carry the same code taxonomy the TEAP uses — matching on the code is exact, explainable and language-independent. A code is compared **number by number**: the TEAP's `DOP.001.02` and the sheet's `DOP.001.002` are the same code, and so are `DOP.001.04-02` and `DOP.001.004-002` (a dash and a dot are the same separator). Such a match is exact, and its reason says the number was written with a different width.

When no room carries the exact code, the matcher looks at the **code family** — the rooms whose code starts with the same letters and first number — and ranks them, in this order:

1. **The name.** A room the sheet calls what the TEAP line calls it (allowing for small differences) comes first: `HOC.200.02` “Umpire Managers” goes to `HOC.200.001` “Umpire Managers” rather than to `HOC.200.002-002` “Umpire Change Room”. When a deeper code exists the row says so (*the name points at …, the code goes deeper at … — check the sheet*).
2. **How deep the code agrees.** `DOP.001.04` against the sheet's `DOP.001.004-001` shares three parts; against `DOP.001.002`, two. The deeper one wins.
3. **Whether the room contains the line or is contained by it.** The family head (`PRS.001` Venue Media Centre, for a `PRS.001.03` Photo Work Room the sheet does not print) or a room the sheet divides comes before a neighbouring code, which is another room for certain.
4. **Whether the room is free.** A room another TEAP line already holds by an exact code comes after a room nobody holds — so `DOP.001.04` “Processing Room” goes to the Processing Room the lines `DOP.001.04-02` and `-03` do not take. A held room is still offered when nothing else fits (its reason says which line holds it): a room may rightly serve several lines.
5. **The sheet**, last: the more detailed one.

Two rooms that none of this tells apart are reported as *ambiguous* and wait for you. A code-family match is **never confirmed on its own**: it may be the same room or the one next door, and it is precisely the row that needs a person. The name is the fallback when no code of the family is on the sheet: synonyms and abbreviations act inside the normalised name and the similarity, not as a step of their own, and the building named on the TEAP line only nudges the confidence when it appears in the room's name.

What the matcher confirms itself — the exact matches. Two answers need no person: an **exact code** with no rival room, and an **identical name** on a TEAP line that carries **no space code**, when only one room on the drawing answers to it. Those rows arrive confirmed, with a green `✓ exact` where the `Confirm` button would be. An identical name on a TEAP line that *does* carry a code the drawing does not print is not confirmed — the name alone may belong to the room next door — and neither is a name or code that two rooms answer to. A project matched before this rule confirms its identical names at the next `Run matching`; your own decisions are kept as always.

CONTROL	WHAT IT DOES
`Run matching`	Runs the matching in the background. When it finishes it reports how many matched, the rate, how many are not confirmed yet and how many found no room — counted **after** your decisions are laid back over the answer — and how many earlier decisions were kept. The notice's no-room count **leaves out** the rows where you picked `— none —`; the `no room found` tile includes them, so the two figures can differ.

Three tiles, and they are the filter

Above the queue there are three figures, and each is a **button**: click it and the queue shows exactly those rows; the pressed tile is marked, the queue says *Showing only*, and a second click (or *show every row*) brings the whole list back. The figures are counted **from the queue's own rows** and never disagree with the table. Before the first `Run matching` the queue is empty and the tiles read 0; the first run fills the queue.

TILE	SHOWS
TEAP spaces	Every row of the study — the TEAP spaces that request ports, with or without a room.
matched	The rows **with a room** on the drawing, whoever chose it (the matcher or you) and confirmed or not. Who decided is written on the row itself: *your choice* (you picked the room or clicked `Confirm`), *exact code, confirmed by the matcher* or *identical name, confirmed by the matcher*, or nothing. The `Confirm` button is shown on every row with a room that nobody has confirmed yet — including a room picked by hand, which already shows ✓*your choice*: choosing is not confirming, and the analysis treats that row as *match not confirmed* until you click `Confirm`.
no room found	The rows **without a room**: the matcher found none, or you picked `— none —` under `Drawing space` (the row says *you said this space is not on this drawing*). It is the same answer, so it is one tile. Choose the room in the select, or leave it with none: the space is pending until you choose its NER on Demand per space, and it shows on Exceptions.

`matched` + `no room found` = `TEAP spaces`, always. The line under the tiles also says how many rows with a room are waiting for your `Confirm` and how many TEAP spaces with no ports were left out of the study (those are in no sum; to see them, use `show TEAP spaces with no ports` on Demand per space). When it finishes, `Run matching` reports how many of your decisions the run found and kept.

The queue, ordered by demand

Worklist — largest demand first. The rows that **still need you** come first — no room chosen, or a room not confirmed — and the rows already settled (confirmed by the matcher or by you, or set to `— none —` by you) follow at the end, a little dimmed. Within each group the queue is **ordered by port demand, largest first**; a click on a column heading sorts the whole list by that column instead. That is deliberate: a space carrying 110 ports deserves an hour of attention, one carrying a single port does not. Work down from the top and stop when the remaining rows stop mattering. The heading says how many rows there are, how many still have no room chosen and how many ports await a decision; rows with no room are flagged on the edge.

COLUMN / CONTROL	SHOWS OR DOES
Ports · TEAP code · TEAP name	The TEAP line: the port demand, the space code and name.
Drawing space SELECT	Opens with *Suggested — closest code and name*: up to six rooms most likely to be this line's, ranked by how many parts of the code agree (number by number) and by name, the matcher's own alternatives nudged up. Under it, *Every room, A–Z*: every drawing space, **alphabetically ordered** with correct numeric ordering (`SPT.6` before `SPT.16`), each with its page (`PRS.001 · Press Workroom (p12)`). The column has a fixed width; a name too long for it is cut, and the whole name shows when the pointer rests on the select. Choosing one **saves immediately** but **does not confirm**: a room picked by hand stays unconfirmed — the analysis reads it as *match not confirmed* — until you click `Confirm`. Changing the room on a row that was confirmed (by you or by the matcher) drops that confirmation: the new room needs your `Confirm`. The `— none —` option says the TEAP space **has no room on this drawing**: the choice is written on the TEAP line, kept on every `Run matching`, and the option then reads `— none (your choice) —`. The space counts in the `no room found` tile and enters the analysis with no room — see A TEAP space the drawing does not show. To hand the row back to the matcher, use `Undo`. The `none` button beside the select sets `— none —` in one click, without scrolling to the top of a list of hundreds of rooms; it is greyed out on a row already set to `— none —`.

COLUMN / CONTROL	SHOWS OR DOES
Page	Which page of the drawing the chosen space is on. It answers “where is this” without leaving the screen — and it is a **button**: one click opens the Drawing viewer on that sheet, zoomed in on the chosen room and ringed in orange. — when no room is chosen.
Confirm	Signs this row off — and only this one. Disabled until a room is chosen (*Choose the room first*). On a row already confirmed there is no button: a green `✓ exact` when the matcher confirmed it (an exact match, see above) and `✓ confirmed` when you did; `Undo` beside it takes your decision back. On a room picked by hand, `Confirm` is what signs it off.
Undo	Shown on every row **you** decided — a room chosen by hand, a confirmation or `— none —` — and takes the decision back: the row returns to **the matcher's own answer**, the same room and the same method `Run matching` would show, confirmed only if the matcher would confirm it (an exact code with no unresolved rival — the same code on a key plan or a less detailed sheet is resolved, not a rival; an exact name in the right building can reach the same bar). Nothing is invented: a row the matcher could not place goes back to *no room*. The notice says where the row went. On a row whose matcher answer is not stored yet (no run has refreshed it): `Undo` takes the decision back, leaves the row with no room and asks you to run the matching, which brings it back. Every `Undo` goes to the audit log with what was taken back.

The `Drawing space` list is **one alphabetical list** of every room on the detail sheets, tied to a TEAP line or not — two TEAP lines can rightly point at the same room, and a room is found by its code or name wherever it stands. It is the only search “on the drawing” there is, and it always starts from a TEAP line. Closed, a row shows only the room it is tied to; the full list is built when the pointer reaches the select or the keyboard moves into it — on a venue with hundreds of rooms, building every list of every row at once is what froze the browser for a moment after `Run matching`. Choosing a room or pressing `Confirm` reloads the rows, not the list of rooms.

A decision that disagrees with an exact code. Your choice is never changed by a run. But when the sheet prints the line's exact code on *another* room — typically a room confirmed before 5.2.2, when `TEC.100.02` and `TEC.100.03` were both proposed `TEC.100-001` — the row's status line turns amber: *⚠ the sheet prints this exact code on another room: … — check your choice*, and the line under the tiles counts those rows. `Undo` hands the row back to the matcher's answer; choosing the room in the select does the same with your own sign-off still to give.

The queue holds still while you work. Its order is taken when the screen opens and kept while you choose rooms and confirm: a row you settle stays where it is, dimmed, instead of moving to the end and pulling the rows below it up under the pointer. `Re-sort`, above the table, puts the rows that still need a look back on top; pressing a tile or `Run matching` does the same. A row that leaves the pressed tile's filter (a room chosen on `no room found`) stays in view until then.

Under the select, on a line of its own that is always there (so a decision never changes the row's height), every row you decided shows `✓ your choice — saved, and kept when the matching re-runs` — on a room picked by hand that says the choice is saved, not that it is confirmed: while `Confirm` is lit it still waits for your sign-off; a row where you picked `— none —` shows *you said this space is not on this drawing — kept when the matching re-runs; Undo takes it back*.

RE-RUNNING NEVER ERASES YOUR WORK

Your answer lives **on the TEAP line itself**, written the moment you give it — choosing in the select, clicking `Confirm`. The matching table is a *derived* table that `Run matching` refreshes row by row and that **reads** your decision rather than trying to rebuild it. Rows nobody has touched still take the fresh answer, or the step would be useless.

One deliberate exception: if the drawing was re-imported, the chosen room no longer exists in the database and the row would be left decided pointing at nothing. That decision is **not** kept — carrying the sign-off onto whatever the matcher picks next would be signing for something you never saw. The row returns to the queue and the notice says how many did (*decision(s) released — the room is no longer on the drawing*).

SIMILAR NAMES ARE NOT A REQUIREMENT

The room list is **not filtered by name similarity**: it carries every space on the drawing. If the TEAP says `OFFIC2` and the sheet says *Observer Office 2*, pick it anyway, then click `Confirm` — no metric was going to connect those, and that is exactly why the choice is yours. If the space is not on this drawing, pick `— none —`: the decision holds, and the space goes on to the analysis with no room, waiting for the NER that serves it. Every choice belongs to this project alone; nothing is carried to another venue.

HOW TO JUDGE THE QUEUE

A smaller queue is not automatically a better one. What matters is *what each remaining row asks you*. Twenty rows saying “the drawing divides this code into …” are twenty two-second checks; forty saying “this code is on four sheets, I don't know which” are forty investigations. The evidence for every match — method, score, confidence, reason — is on the *Matching* sheet of the analysis workbook and in the audit log.

STEP 6

Demand per space

How many network ports each space demands, **which NER serves each one**, at what distance, with what status — and the button that produces that answer, on the same screen.

The serving-room rule

1. **The TEAP names the room.** When the imported sheet carries a serving-NER column (mapped on Import as `serving ner`) and the name matches a NER on the list, there is no doubt: the row shows `TEAP: …` and *named in the TEAP — no doubt*. It is resolved by name (*NER 03* finds *NER 3*; an earlier name still finds a renamed room). If no NER carries that name, the row warns in amber and moves on to the next rule.
2. **You chose.** The `Served by` select on the row, with a justification recorded in the audit log. It outranks the TEAP when both exist — you are looking at the drawing and the

TEAP is not — and the row says which one applied.

3. **The nearest NER within the limit that still has free ports.** If the nearest is full, the next one. If *every* room within reach is full, **the nearest anyway** — with status *Insufficient capacity*, the ports counted in that NER's load (utilisation goes past 100%) and the row under Exceptions. The overload is reported, not hidden: the decision to draw a second NER there is made on the report.
4. **No NER within 90 / 70 m.** Nothing serves it. The row says — *none within 90 m · nearest: NER 5 at 120 m*, status *Out of limit*, and the ports go on **nobody's** load — the cable does not reach.

Whichever leg chose the room, **distance, limit and capacity are still evaluated and reported**: a chosen room 200 m away comes back *Out of limit*; one with no ports free comes back *Insufficient capacity*; and both land under Exceptions like any other. Letting the choice silence those checks would turn the list into a way of laundering a violation — and would destroy the use case: you choose *because* the other room is overloaded, so the tool has to keep saying so.

A room the matcher found is served before you confirm it. When the TEAP × drawing match of a space is not confirmed yet — the matcher placed it on a room and nobody pressed `Confirm`, or the same code is on two sheets — the space is still measured from that room and **served by the nearest NER** under the usual rule: the status is the distance's (*Served*, *near the limit*, *Out of limit*), its ports count on that NER, and it does not go to Exceptions. What stays is the warning: the `Matched to` column says *room not confirmed*, the counter **`served — room not confirmed`** above the table lists those rows (click it to filter), the PDF keeps the orange rectangle around the space and the workbook says *yes* under `Needs validation`. If the room is wrong, the NER is wrong too — confirm or correct the room on TEAP × drawing, then `Run analysis`.

A space whose page scale is not confirmed yet gets the status *scale not confirmed*. With no NER fixed, it is **not assigned** — that result stays blocked on purpose — and the row shows the nearest room and its distance beside it, rather than a dash. With the NER fixed (chosen by you under `Served by` or named by the TEAP), the space **is assigned** to it and its ports count on that NER's load, but the status stays *scale not confirmed* until the scale is confirmed. A page with **no scale at all** measures nothing: there the space comes back *No NER reaches it*, not *scale not confirmed*.

A NER ON ANOTHER PAGE

A space on level 2 may be served by the NER directly below it on level 1, through a shaft — or by the NER on the neighbouring sheet of the same level. The automatic rule **never** assumes that — the app does not cross pages on its own — but your choice holds: **choosing in `Served by` a NER that sits on another page is itself the statement** that it serves the space (you are looking at the design, the levels and the shaft; the app is not). On the next run that NER serves the space, the ports **count against its capacity**, and the status is *Other level — not measured*, in the *Served — not measured* group: no distance is invented, because two sheets share no coordinates. The row says so in green: *other page — NER x is on page …; this space is on page … · counted on this NER, distance not measured* (the green line appears on the rooms you chose; a NER the TEAP names on another page gets the same treatment in the run, with the row reading *named in the TEAP — no doubt*). The distance is measured once the vertical route (the `vertical` tool of the Drawing viewer) is registered between the two pages — and a traced route (`route`) reaches the shaft from the space and from the NER: the loose end between a point and the nearest traced node is about 60 drawing units, which the scale turns into metres; a shaft on its own, far from everything, measures nothing — and until then the space stays under Exceptions as a reminder, not as a failure. To withdraw the statement, use `× clear my choice` under `Served by` . A NER the TEAP names on another page is honoured the same way.

A NER chosen on the **same page** as the space is measured like any other: too far, the row comes back *Out of limit*. Choosing picks the room, not the verdict.

A TEAP SPACE THE DRAWING DOES NOT SHOW

Every TEAP space that requests a network port has to be served by some NER — whether or not it is found on the drawing. A space with **no room on the drawing** (the matcher did not find it, nobody tied it by hand, or you picked `— none —` on TEAP × drawing) enters the analysis all the same, as *Pending — needs a room or a NER* with the reason *no room on the drawing*: its ports are on nobody's load, and the list says so. Two things resolve it: tying it to a room on TEAP × drawing, once the sheet shows it; or **choosing the NER under `Served by`** , from your knowledge of the venue. Once chosen, that NER serves the space from the next run: the ports **count against its capacity** (and a full room comes back *Insufficient capacity*), no distance is measured — there is no position to measure from —, the status is *Served — not measured* with the reason *no room on the drawing — NER chosen by you*, and the space stays under Exceptions as a reminder to validate the route on site. It is the same treatment as a NER on another page: with no room there is nothing to measure from.

The same space on more than one page

A design book cuts one level into several sheets, and a room on the cut line is labelled on both. The TEAP line is tied to **one** of those labels; if that one is in the corner of a sheet with no NER, the space would come back *No NER reaches it* with the NER that serves it a few metres away on the next sheet. So for every TEAP space already tied to a room the application looks for the **other labels of the same room on other pages** — by the room's full code; with no code, by its name — and shows each one under the space's name:

ON THE ROW	MEANS
✓ also on page N — same code, same level	**Proof.** Same full code and both pages on the same level, read from the title block. Linked automatically, with the evidence kept (hover over it). `not the same space` undoes it if you disagree.
also on page N — same code · same level by the drawing number also on page N — same code · level not read also on page N — same name, no code · level not read	**Question.** The level of one of the pages was not read off the title block — the row then states the level the *drawing number* implies (`VEE-440B5-VVO-O` → LEVEL 04), which is weaker evidence than the title block and so never enough to link by itself — or only the name matches ("Storage" is on every level of every venue). Nothing is measured from there until you answer `Same space` or `Not the same` — one row at a time, with no bulk action, because a wrong "same space" moves a room's ports onto a NER that does not reach it. When the answer would change the result, the row says what it would give: *NER 20 is 34 m from it there*.
× page N — not the same space (your answer)	You said it is not. That page is not used for this row, and re-running the matching does not change that. `undo` hands the question back to the rule.

Levels that are known and **different** produce no link at all: that is another floor, and another floor is declared by choosing the NER of the other page under `Served by`, never inferred. Once a link is confirmed, the analysis measures from **every** label of the space and keeps the shortest real distance — measured on the NER's sheet, at that sheet's scale. The `Distance` column says *measured on page N*, the notes explain why, and the marked-up PDF draws the space and its line to the NER on that page too. Because the point measured is the label position, which may sit at the edge of the sheet, the distance is an estimate — the note says so. The `same space on another page? — to confirm` tile counts the open questions.

Controls and tiles

`show TEAP spaces with no ports (n)`, off by default, brings back the TEAP spaces that request no port — the number in brackets says how many were left out. They show marked *no ports requested in the TEAP — not part of the study* and stay out of the analysis. See The TEAP rules.

`Run analysis` runs the As-Is analysis in the background and reloads the list; beside it, the filter (`all spaces`, one of the four groups, or — under each group — a precise reason from the last run, with its count) and the time of the last run (*Last run*) — or *No analysis yet — the rows show demand only*. When something that changes the result happened after the last run — a NER deleted, moved, renamed or restored, a match decided, a TEAP imported — an amber pill beside it says *changed since the last run (n)*, and the same pill is on Exceptions; the rows still show the last run until you press `Run analysis`. Tiles: `spaces`, `ports`, the **four groups** — `Served`, `Out of limit / over capacity`, `Served — not measured`, `Pending — needs a room or a NER` (see the coverage statuses; each tile is also the filter for its rows, click again for all; each row's group comes from the last run, with three exceptions that move the tile before the next one: a space with a room but no run yet is *Pending* (*not analysed yet*); a space with no room and no NER is

Pending; a space with no room and a NER chosen is *Served — not measured*) —, `photographer areas`, `rooms chosen by you`, `wireless points` (when any are drawn), `same space on another page? — to confirm` (open questions, see below), `tied to another room` (spaces tied to a drawing room with a different code — the `Matched to` column) and — when the TEAP names rooms — `rooms named by the TEAP` (resolved / named).

Nearly every counter is a filter. `spaces` and `ports` are plain figures; `Wi-Fi points →` opens the Wi-Fi step, where they are listed. `near the limit (counted as served)`, `photographer areas`, `rooms chosen by you`, `tied to another room`, `same space on another page? — to confirm` and `rooms named by the TEAP` are buttons, like the four group tiles: click to show only those rows, click again for all. With one of them pressed, the filter select reads `filtered by a counter below (n)` — except `near the limit`, which is the *Near the limit* reason under the *Served* group and reads as such.

The search. The `search code, name, room or NER` box, beside the filter, shows only the rows whose code, name, room on the drawing (`Matched to`), FA or NER contain **every** word typed, ignoring case and accents; the number of rows found shows beside it. It adds to the filter and the counters. The `Demand` buttons on Exceptions open this screen with the search already filled in. The `near the limit (counted as served)` counter, when there are any, shows the spaces served within the limit but close to it — they count in *Served*. It is counted from the last run, not from the rows as they stand now.

The table has a working width: on a smaller monitor it **scrolls sideways**, with the scrollbar mirrored right under the headings (and the usual one at the foot of the box), instead of crushing the columns.

COLUMN	SHOWS OR CONTROLS
Code / Space / FA / Ports	The TEAP space, its functional area and its port demand; the drawing page appears under the name, with the drawing number read off the title block (`page 13 · VFF-410B8-CSL-0`) — and it is a **button**: one click opens the Drawing viewer on that sheet, centred on the room and with a ring round it, as on TEAP × drawing. The pages of the same space on other sheets (*also on …*), the `Page` of each Wi-Fi point and of each NER under *NER load and capacity* are buttons the same way.
Matched to	**The from → to.** The drawing room this TEAP line was tied to on TEAP × drawing. When that room's code is a different one (`PRS.001.03 → PRS.001.02` , or `PRS.006.02 → PRS.006`) the row shows the arrow in blue, the room's name and how it was tied — *tied by hand*, *matched, confirmed* or *matched, not confirmed*. Same code: *same room*. Nothing: *— not matched —*. Sorting by this column brings the redirected ones first.
Demand type SELECT	`general` or `photographer` , with the project's limit beside it. Sets which limit applies to this space; changing it saves at once and applies on the next run.

COLUMN	SHOWS OR CONTROLS
Served by SELECT	The serving NER. The greyed text is the rule's answer (`TEAP: …` , `nearest: …` , or why there is none); choosing a room from the list overrides it and asks for the reason (*Why should this room serve this space? Recorded in the audit log*). Below it: *chosen by you* with the reason, *named in the TEAP — no doubt*, or the warning that the TEAP names a NER that is not on the list. Only NERs are offered — a critical one carries ★. If the chosen NER is on **another page**, the choice stands as the statement: the row says in green which page each one is on and that the ports count on that NER with no distance measured; see the note above. On a space with **no room on the drawing** the greyed text reads *— no room on the drawing: choose the NER that serves it —*, and the room chosen serves the space with no distance measured — see the note above. With a NER chosen by hand, the first option of the list undoes the choice and says where the row goes back to: `× clear my choice — back to: no room on the drawing` (a space with no room), `× clear my choice — back to the nearest NER rule` (a space with a room) or `× clear my choice — back to the TEAP's room: X` (the TEAP names one). Picking it releases the choice, recorded in the audit log.
Distance	The distance to that NER against the limit (`58.2 m / 90 m`) and, where it applies, *straight line*: a preliminary straight-line distance, to be validated against the real cable route. A room on another page with no vertical route: *other level — not measured*; with one registered: the route distance and *via vertical route*.
Status	The **group** the space is in — *Served*, *Out of limit / over capacity*, *Served — not measured*, *Pending — needs a room or a NER* — and, on the small line under it, the precise reason (*within the limit*, *near the limit*, *match not confirmed*, *scale not confirmed*, *no room on the drawing*…). See the coverage statuses.
Notes	The analysis's assumptions and warnings plus the import divergences. Closed, the cell shows two lines and `read (n)` ; clicking opens every note in full, one per line, nothing clipped, and `fold` closes it. `open all` / `fold all` in the heading does it for the whole table. Three sentences repeat on nearly every row (estimated service point, slack factor 1.0, straight line): they are assumptions of the run, not findings about this row.

NER load and capacity

Collapsed under the list; click the heading to open it. Per room, from the last run: `NER` , `Page` , `Critical` , `Spaces` , `Wi-Fi` (the Wi-Fi points it reaches), `Ports assigned` (the TEAP ports of the spaces it serves, the `DWISA` lines included; a Wi-Fi point adds none), `Total ports` (or `unknown`) and `Utilisation` — `–` when capacity is unknown, never 0%; red, with *over capacity*, past 100%.

Criticality is not a control here: once the room is chosen, the area inherits that room's criticality, which comes from the NERs screen. The rule that prevents double counting is the usual one: trust the TEAP line's explicit total when it exists, rather than recomputing from the catalogue — see the four demand rules.

STEP 7

Wi-Fi

The spaces the TEAP says have wireless, marked on the drawing beside their rooms, and every Wi-Fi point with the NER that serves it. It comes after TEAP × drawing because a point goes beside the room its space is tied to: settle the rooms first, then place.

Three buttons sit on the first card: `Place Wi-Fi from the TEAP` , `Spread stacked Wi-Fi points` and `Run analysis` — the same analysis as on Demand per space, here because a Wi-Fi point is measured to its NER from where it is drawn, so its

distance and status change only when the analysis runs again. The line beside them says how many points sit on top of another one and when the last analysis ran.

When a space's room changes after its point was placed — a wrong room corrected on TEAP × drawing — press `Place Wi-Fi from the TEAP` again. A point this step placed, and that nobody dragged, **moves to the room the space is tied to now**, beside its label (a point that already sits beside any label of that room, on any sheet, stays where it is); the move is in the audit log. A point you dragged stays where you put it, and the notice names it (*you dragged … sit away from their space's room — kept where you put them, check them*). Then run the analysis.

Place Wi-Fi from the TEAP

The first card of the screen, for the spaces the TEAP already says have wireless. `Place Wi-Fi from the TEAP` marks them where their rooms are: for every TEAP space in scope with a wireless line (the `DWISA` line — *Wireless Connection - Shared Internet Access (SIA) Wireless LAN*) and whose room is on a drawing — matched by the matcher or tied by hand on TEAP × drawing, not rejected — it places **one Wi-Fi point**, whatever the quantity on that line, a little below and to the right of the room's label, numbered in the project's `WIFI-1`, `WIFI-2` … sequence, and **tied to that space** (`TEAP space` already filled in; the Inspector says *placed from the TEAP*). **A Wi-Fi point says the space has wireless; it is not an access-point count.** The `DWISA` quantity stays on the TEAP line, and in the workbook, as the space's wireless ports; the number of access points comes from the wireless design. **The point is not a TEAP product and adds no port:** the space's `DWISA` line stays its wireless port, counted on the space's row and on the NER that serves the space, whether the point is drawn or not. Drag it to where the space's wireless is wanted, and its distance to the NER is checked from there. A point placed this way keeps its space wherever you drag it; change or clear the space in the Inspector if it belongs to another one. Never on a key plan. **A space with no room on the drawing gets nothing** and stays *Pending — no NER*: tie its room first. Several TEAP lines tied to the same room — six photo positions on one press room — get their points **side by side**, in rows of four beside the label, never one on top of another, and far enough apart that their names do not touch on the marked-up PDF; a slot already holding a Wi-Fi point is skipped. **Running it again never duplicates:** a space that already has its point is left alone; one with none gets it — unless **you deleted its point**: a deleted Wi-Fi point stays deleted, the space stays without one and the notice counts it (*space(s) whose Wi-Fi point you deleted*); drawing a point with `+Wi-Fi` for that space, or tying one to it in the Inspector, brings it back. A point deleted with a version before 4.8.0 was not recorded with its space, so the first run after the update places it again and its name is in the notice; delete it once more and it stays deleted; one with more than one loses the extra points this step placed (the newest first) — a point you drew by hand is never removed, only counted in the notice. A second press while it is still running is refused. It asks before placing; the notice says how many points went into how many spaces, how many spaces already had theirs, how many extra points were taken back, how many points drawn by hand go beyond one per space (kept — check them) and how many spaces have no room yet. Check them in the Drawing viewer; a point placed by mistake is deleted there or in the list below like any other.

`Spread stacked Wi-Fi points`, beside it, is for points that sit exactly on top of each other — what a version before 4.8.0 drew for several TEAP lines of one room, and what is easy to miss on the sheet and unreadable on the marked-up PDF. The line beside the button says how many points sit on another one, in how many stacks; the button asks first,

then keeps the oldest point of each stack where it is and moves the others beside it, in rows of four. Each point keeps its space, the NER chosen for it and indoor/outdoor; every move is in the audit log. Run the analysis afterwards: the points are measured from where they are now.

Wireless points

The Wi-Fi points — placed from the TEAP on this screen (one per space with wireless) or drawn in the Drawing viewer with `+Wi-Fi` — are listed in the second card, with a search box above the table: `Wireless point`, `Type` (*Wi-Fi*), `Indoor / outdoor`, `Page`, `TEAP space` (with how the tie was made: *proposed*, *tied by you* or *from the TEAP*), `Served by`, `Distance`, `Status` and, at the end of the row, `Delete` for a point placed by mistake (it leaves the list at once). **A Wi-Fi point marks a space that has wireless; it is not an access-point count and takes no port.** A point tied to a TEAP space follows **the NER that serves that space** — the one the space's row shows under `Served by`, whether you chose it, the TEAP names it or it is the nearest — and is measured from it under the general limit; that is what keeps the point and the space's `DWISA` port on the same NER. If the space has no NER yet, the point waits with it (*No NER reaches it*, with the reason, and a row under Exceptions) instead of going to whichever NER is nearest the point. A point with no TEAP space goes to the nearest NER within the limit; with nothing within the limit it gets no room — the nearest and its distance beside it, and a row under Exceptions.
`Indoor / outdoor` says where the service will be given — `Indoor` for a covered area, `Outdoor` for an open one, or *not said* — so the LLD team knows the setting each access point is proposed for. It is said by you, never read off the drawing, takes effect at once (no re-run: the workbook reads it off the point when it is exported) and is the `Indoor / outdoor` column of the *Wireless points* sheet, beside `Type`, with three *Totals* lines on the *Summary*. The same select is in the viewer's Inspector. The `Wi-Fi` column of *NER load and capacity* counts the Wi-Fi points each NER reaches; they are not spaces, so they are not in `spaces`, `ports`, `within limits` or `not covered`.

The NER of each point (the `Served by` column, a select). Three sources, in this order: **the NER you chose for the point** (the row says *chosen by you*); otherwise **the NER fixed for the TEAP space the point belongs to** — chosen by you under that space's `Served by`, or named by the TEAP — and the select reads `as its TEAP space: …` with the line *the NER of its TEAP space*; otherwise the nearest-NER rule above. So the Wi-Fi point of a room served by a NER on another page follows that same NER, counted on it with the distance *not measured* (*Other level — not measured*), with nothing to do. With a choice of yours, the first option of the select is `× clear my choice` and says what the point goes back to. The ports do not change with the choice: the wireless port is the space's `DWISA` line and stays on the NER that serves the space; only the NER the point is measured to changes. By the rule, with no choice, a point is never served by a NER on another page.

THE TEAP'S WIRELESS AND THE WI-FI POINTS DRAWN

The TEAP requests a space's wireless as a service line (*Wireless Connection - Shared Internet Access (SIA) Wireless LAN*, code `DWISA`), usually with **one port**: the access point's uplink. **The TEAP is the only source of ports.** That line's ports enter the space's total like any other line and land on the NER that serves the space — whether or not a Wi-Fi point is drawn for it. A Wi-Fi point drawn on a sheet is **not a TEAP product** and **not an access-point count**: it marks the space as having wireless, never a port of its own. So five points drawn by hand for a space whose `DWISA` line asks for one still put one wireless port on the switch; if the space really needs five, its `DWISA` quantity is what has to say five, and the number of access points comes from the wireless design. The space's row says under `Notes` whether a Wi-Fi point is drawn for it.

The tie between the point and the space is **proposed** on placing (the nearest room on the same sheet that is matched to the TEAP, up to the general limit) and shown in the viewer's Inspector and in this table's `TEAP space` column, with *proposed*, *tied by you* or *from the TEAP*. Change it in the Inspector when the point belongs to another space; clear it when it belongs to none (it is then outside the TEAP and counts no port). The proposal follows the point when it is dragged; your own choice and a tie made from the TEAP do not.

STEP 8

Exceptions

Every TEAP space that still needs you, in one list, with the NER changed right there. It is the to-do list before exporting anything to anyone else.

The list is made **from the same rows as Demand per space**: the last saved analysis plus the choices made since. The two screens never disagree. Every space that requests ports and is not simply served within the limit is on it — including those with no room on the drawing and no NER yet — and every wireless point with a problem.

CONTROL	WHAT IT DOES
Re-run the analysis	Runs the analysis again (in the background, with the progress bar) and reloads the list. It is the same analysis as `Run analysis` on Demand per space. *Last run* says when the last one was.
all that need you	Tile with the total. The other four filter the list (click again to see everything): `Pending — needs a room or a NER` (no room and no NER, waiting for the match or the scale to be confirmed, *no NER reaches it* or *not analysed yet* — and the pending wireless points), `Out of limit / over capacity`, `near the limit` (served, within the limit but close to it) and `Served — not measured` (a NER on another page, or a space with no room and a NER chosen).
Served by SELECT	The same list of NERs as on Demand per space: choose the NER that serves the space (it asks for the reason, written to the audit log) or `× clear my choice — back to the rule` to release it. The greyed text is the rule's answer (`nearest: …` — also when that NER is the one the TEAP names —, `TEAP: …` when no run has served it yet, or `— no NER yet: choose one —`). On a wireless point it chooses that point's NER.
Distance	The distance to the serving NER and the limit; on a near-limit space, the margin left. With no NER, the nearest one and how far it is.

CONTROL	WHAT IT DOES
Other NERs measured	Up to three other NERs the analysis measured from the space, nearest first, the serving one left out, each with its distance — green within the limit, red outside it. `use` beside one first asks for the reason (written to the audit log) and then chooses it for the space. Only NERs on the same page (or joined by a registered shaft) are measured; a NER on another page is chosen under `Served by` and stays *not measured*. When the next NER is already out of the limit there is no closer room today: the way out is a new NER on the drawing.
Demand · Drawing	`Demand` opens Demand per space already filtered to this space (the search box is filled in with its code). The search finds the code inside other codes too: `PRS.006` also brings up `PRS.006.02`. `Drawing` opens the Drawing viewer on the room's page, centred and ringed; on a wireless point it finds the point.

The list deals only with TEAP spaces **that request ports** and with wireless points. There is no list of “rooms on the drawing not in the TEAP”: the check runs from the TEAP to the drawing, and a room no TEAP line asks for is not an exception. The table sorts by any column (see Sorting the tables); by default pending rows come first and, within each group, the largest demand first.

STEP 9

Reports

The deliverables. All in English whatever the interface language; the marked-up PDF and the workbook are stamped with the version of the application that produced them (the CSV carries data only).

CONTROL	WHAT IT DOES
Marked-up PDF	Every drawing book of the project, in the order they were uploaded, page by page in one file, with the markers, the areas coloured by status, the lines from each space to the NER serving it, the legend on the first page and the disclaimer on every page. It runs in the background — on a large venue it takes a minute or two; the bar names the page it is on.
with legend CHECKBOX	Turns the legend box on and off. On, the box is drawn on the **first page of the file** only (in a project with several books, the first book's first page); every other page carries the short footnote at its foot — the limits, the slack, the scale, the “no coverage analysis” warning (when it applies), the disclaimer and the version — and the line *Legend: see the first page of this file.* Off, there is no box anywhere and no colour swatches — **not** the statements: the footnote is on every page, without that line.
with NER → space lines CHECKBOX	Turns the dotted lines from each space to the NER serving it — with the distance written on the lines from spaces; a wireless point's line carries none — on and off. Off, nothing else changes: markers, area colours and legend stay as they are. A space served by a NER on *another page* has no line to draw — with the lines on, it says beside it which room serves it and on which page (`NER 19 (p. 12)`).
Composite map PDF	One page per composite map built: a level of a book on one map, every NER with its reach, the legend on the first page. Runs in the background.
Analysis workbook (XLSX)	The full analysis as a workbook (sheets below). Immediate.

CONTROL	WHAT IT DOES
Coverage CSV	The coverage as CSV, one row per space and per drawn Wi-Fi point (a Wi-Fi point always with 0 ports): key, name, page, demand type, ports, NER, the NER's internal id, who chose it (`you` / `the engine`), critical, distance, method, limit, margin, status and `needs_validation` (scale, route or match only; an assumed port count is flagged only in the workbook). Immediate.

The generated files appear as links under the buttons and are kept under `data\exports\project_<id>\`. Each export **replaces** the previous file of the same format (the name is the project's and the scenario's): exporting with and without the legend keeps only the last one; copy it aside first if you want both.

Every sheet of the workbook arrives with the **filter on** in its heading row and the first row frozen: filter by NER, by status or by `Needs validation` with nothing to set up.

What the marked-up PDF draws

- **A clean copy of the book.** The reviewers' own markups — the PDF *annotations* a comments round leaves on a sheet: the red *COW #1 / NER 14* free-text boxes, the ink blobs, the sticky notes — are taken off the copy before anything is drawn on it. They are not part of the drawing (the design team's linework is page content, and stays), and the application draws its own markers in their place. The book on disk is never touched, and the viewer still shows them.
- **Every room with a position**, serving anyone or not — NER, TER, IDF, COW and DIVA node — on **every sheet it is drawn on, where that sheet draws it** (a NER drawn on a level plan and on two details appears on the three, each time at its own marker, with its circles and the lines to that sheet's spaces), as 24 pt dots in the kind's colour, ringed in black, with the name in **bold** on a translucent white tag *under* the dot (under, not beside — a habit from the days the reviewers' annotations were still on the copy and every viewer painted them above the content). Rooms with **no position** stay out: there is nowhere to draw them, and a marker invented for completeness is a lie on a construction document.
- **The spaces** filled in the colour of their status (green, orange, red or grey) and an orange rectangle around those whose match is not confirmed.
- **The Wi-Fi points**, with the Wi-Fi sign in light blue, the name (`WIFI-3`) under it in bold, at the NERs' size, in dark blue on a white tag so it reads over the linework, a disc in the status colour under the sign and — with the lines on — the line to the NER it is measured to (no port of its own: the port is its space's `DWISA` line).
- **A space measured on another page** (the same space labelled on two sheets) is drawn on both: on the sheet the TEAP line is tied to, with the NER's name and page beside it; and on the NER's sheet, with the line, the distance and the note *TEAP line tied on p. N* (with several series, the sheet's drawing number).
- **The reach circles** around every NER (and every NER reference), to the sheet's scale: dashed blue for the general limit and dotted purple for photographers, radius = limit ÷ slack. A sheet with no scale gets none. And the **named areas** and the key-plan **references**, as in the viewer.
- **The legend**, at the top left of the **first page** of the file, with the limits, the slack, the scale, the disclaimer and *NER Spatial Coverage Analyzer 5.2.2*; its wireless row reads *Wi-Fi - space with wireless*. Every other page carries the same statements as small

lines along the foot of the sheet, with *Legend: see the first page of this file.* With `with legend` off, every page carries the footnote and none carries the box.

Two different documents come out of the same button, at two different moments. Late in a study it is the analysis: rooms, coverage, distances. Early on it is the drawing — *"here are the racks I am proposing, put them in front of the design team"* — and at that point there may be no TEAP and no matching. When the project has no analysis at all yet, the sheet **says so**, in the legend or the footnote, in amber: *No coverage analysis on this sheet…* (a decision per project: a key plan with no space analysed on it, in a project that already has an analysis, gets no note). Silence there would be read as "everything is covered". Only the PDF does this: the workbook and the CSV always carry an analysis — with none saved, or with one that something changed after (a NER deleted, moved or renamed, a match decided, a TEAP imported), they run it first and save it, and the notice at the end says so (*the analysis was run again first: n change(s) after the last run*). Without a TEAP they return the error.

The saved analysis records *which* space, never *where*: if a space has been unmatched from the drawing since the analysis was saved, it is not drawn at an invented position — it drops out of the PDF and appears under Exceptions with the reason.

The workbook's sheets

Wherever a sheet is named — the `Page` columns, *Scales*, the exceptions — a project with several series names it by its drawing number and series page (`VAA-411B3-VBV-O (400 p.6)`); a project with one book keeps the page number.

In this order — the three the network team works from first, *TEAP by NER* with a **red** tab and *NERs and ports* and *Wireless points* with a **yellow** one, then the rest with no colour: *TEAP by NER*, *NERs and ports*, *Wireless points*, *Summary*, *TEAP and coverage*, *Matching*, *Match exceptions*, *Exceptions*, *TEAP spaces with no ports*, *Scales*, *Coverage figures*. The file opens on *TEAP by NER*. A sheet with nothing to say (no wireless points, no space without ports) is not written; the others keep their places.

> ONE SET OF NUMBERS
>
> Every sheet counts the same ports and the same NERs. *Total ports requested (TEAP)* on the *Summary*, *ports requested by the study (TEAP)* on *Coverage figures*, the sum of `Counted ports` on *TEAP and coverage* and the *Grand total* of *TEAP by NER* are one number; *Ports on a NER*, *ports on a NER*, the `TOTAL` of `Ports assigned` on *NERs and ports* and the NER groups of *TEAP by NER* are another; the ports pending are the difference. Indoor / outdoor counts the same rooms everywhere — the NERs of the last run; a NER taken out of the register with Remove on the NERs screen is never counted. All of them come from the last saved analysis: after a change, run the analysis again before exporting.

SHEET	WHAT IT CARRIES
TEAP by NER	**The summary by NER.** The previous sheet's lines, summed the way the LLD team reads them: `NER` → space → product, with a total per space, one per NER and a *Grand total* at the end. Seven columns: `NER`, `Critical`, `Space code`, `Space name`, `Product code`, `Service / product`, `Ports` (the counted ports). **Only TEAP lines**: a Wi-Fi point is not a TEAP product and is never listed; a space's wireless is its own `DWISA` line, under the space like any other product. So each NER total equals `Ports assigned` on *NERs and ports*, and the *Grand total* equals *Total ports requested (TEAP)* on the *Summary*. The lines with no NER form one group, *Pending — no NER*, last. Lines counting 0 ports (monitors, phones, TVs) are left out, and how many is written under the total. The totals are `SUBTOTAL` formulas: nothing is counted twice, and a filter narrows the *Grand total* with it. Excel's **1 / 2 / 3** buttons, on the left, collapse the sheet to NER totals, space totals, or show every line. It is not a PivotTable: fields cannot be dragged — for another cut, build one on *TEAP and coverage*.
NERs and ports	One row per NER — and only per NER: `NER`; `Full name` — the project's venue code in front of the NER name (`VHH_NERc_01.1`); with no venue code on the project, the name alone; `Space code` — the room's code as the AAM or the VDB writes it (`TEC.010-001`), from the NERs screen; `Page`, `Critical`, `Indoor / outdoor` (as said on the NERs screen; blank when not said), `Spaces served`, `Ports assigned`, `Total ports`, `Utilisation %`, `Wireless points` (the Wi-Fi points drawn that the NER reaches — a count, no ports), `of which wireless ports` (the `DWISA` ports of the spaces it serves, part of `Ports assigned`) — the two are counted from different sources on purpose and check each other: a point follows the NER of its space, so they differ where a space's `DWISA` line asks for more than one port (one point marks it), where a space with wireless has no point drawn, or where a point was drawn with no TEAP space, `Fibre runs to TER` — the fibre runs from the TER the room needs, by the project's rule (`2` per critical NER, `1` per non-critical one, under Project rules) — and `Notes`, what was written under `Notes` on the NERs screen. `Utilisation %` is a **formula** (`Ports assigned / Total ports`): whoever fills in or corrects `Total ports` on the sheet itself sees the figure follow. A blank `Total ports` means *unknown*, the utilisation stays blank, and it is never read as zero. A blank row under the NERs sets off the `TOTAL` row: live sums (`SUM`) of spaces, ports, wireless points and fibre runs, and the indoor / outdoor count (*n indoor / m outdoor / k not said*), outside the filter range, so sorting or filtering the rooms never drags it into the data.
Wireless points	Only when Wi-Fi points are drawn: `Wireless point`, `Type` (`Wi-Fi`), `Indoor / outdoor` (`indoor` — a covered area — or `outdoor` — an open area — as said on *Demand per space* or in the viewer's Inspector; blank when not said), `Page`, `Serving NER`, `Distance (m)`, `Status`, `TEAP space`, `Tie` (*proposed* / *by you* / *placed from the TEAP*) and `Notes`. No ports column: a Wi-Fi point has no port of its own — its space's `DWISA` line is the port. The NER and the distance say whether the 90 m rule holds where the point is drawn; a point tied to no TEAP space says so (*not a TEAP product, no port counted*).
Summary	Project, *Generated* (the time it was produced), *Generated by* (the application and its version) and, right under them, **the totals** (*Totals:* rows): total ports requested — **the TEAP's ports**, and of which wireless (`DWISA`) —, ports on a NER, ports on no NER yet (pending; the last two add up to the first), NERs on the drawing (critical / non-critical), **fibre runs to the TER** in total and by criticality (*n critical × 2*, *m non-critical × 1*), *Wi-Fi points drawn (one per space; not an AP count)*, the NERs said to be **indoor**, **outdoor** and not said, and the same three counts for the Wi-Fi points. Then the limits in force (with the *Near-limit threshold* and whether cross-floor service without a route is allowed, *Cross-floor without a route allowed*), the slack factor, the fibre rule per NER, the source files, the summarised result, the legend for the four demand rules and the disclaimer.
TEAP and coverage	**The main sheet.** Seventeen columns: `Space code`, `Space name`, `Building`, `Level`, `Page`, `TEAP row`, `Product code`, `Service / product`, `Qty`, `Ports` — the TEAP as it arrived — and on the right the answer: `Serving NER`, `NER critical`, `Distance (m)`, `Status` (the group), `Status detail` (the precise reason), `Needs validation` and `Counted ports` — what the analysis counted for the line: always the same as `Ports`, the `DWISA` wireless line included, since only the TEAP carries ports. One row per service, and nothing else — no subtotal rows, so the sheet can be sorted, filtered or pivoted without cleaning. A space with no TEAP lines still gets one row, so it cannot vanish from the report.

SHEET	WHAT IT CARRIES
Matching · Match exceptions	The matching as the project holds it — the matcher's answer with every human decision laid over it — row by row: TEAP code and name, drawing code, name and page, method, score, confidence, reason, confirmed, by whom (*the matcher* or the user), notes. And the matching exceptions, always starting from the TEAP: TEAP lines with no room, rejected lines, codes of TEAP-tied rooms that appear on several sheets, pairs awaiting validation. Rooms on the drawing that no TEAP line asks for are not listed.
Exceptions	The spaces with a problem — no NER reaches them, out of limit or out of capacity —, those near the limit, and those served with no distance measured (another level, no room on the drawing), as a reminder, with `Space key`, `Space name`, `Page`, `Status` (the group), `Status detail` (the status code, e.g. `near_limit`) and `Detail`. A space with no room on the drawing and no NER chosen gets no row here: it is already on *TEAP and coverage* as pending and under Exceptions on screen.
TEAP spaces with no ports	Only when there are any: the TEAP spaces left out of the analysis because they request no port: `Space code`, `Space name`, `TEAP lines`, `Why it is not analysed`. The main sheet carries only the spaces under study; this one exists so that nothing leaves the workbook without a trace.
Scales	The scale used on each page, where it came from, the metres per PDF unit, and whether it is confirmed.
Coverage figures	**The percentages.** Six columns — `Section`, `Figure`, `Count`, `Of`, `Share`, `Note` — and one row per figure: how many spaces the TEAP has and how many request a port (the study); how many were **found on the drawing** (confirmed by you, confirmed by the matcher, matched but not confirmed, **tied by hand**) and how many were **not** (of which served by a NER you chose, of which rejected); the last run's coverage by group and by reason; the ports requested (the TEAP's, and of which wireless), on a NER and pending — the same numbers as the *Summary*; the NERs of the register (drawn by you, from the drawing, with a capacity, over capacity, serving nothing, on the drawing in the last run, indoor / outdoor / not said, and the **fibre runs to the TER** by the project's rule); and the section *Wi-Fi points drawn* (tied to a TEAP space or to none, within reach of a NER or not). `Share` is a real number formatted as a percentage, over the denominator in `Of` — it charts directly.

Serving NER, NER critical and Needs validation

`Serving NER` carries the **name** of the room serving the space and `NER critical` says whether it is critical. They are the two columns this file exists to add to the TEAP. `Needs validation` answers several doubts at once — a space not analysed yet, an unconfirmed page scale or an unvalidated route, an unvalidated TEAP × drawing match, **or** a port count not read off the TEAP (rules B, C and D below). Filter it to `yes` and you have the whole list of what a person still has to check.

The four demand rules

For every TEAP line **exactly one** of four port-demand rules fires — that is what structurally prevents double counting.

RULE	WHEN IT FIRES	WHAT IT DOES
A	the TEAP line already carries a port count	uses the number verbatim, and reconciles it against the catalogue
B	it does not, but the product is in the catalogue	quantity × ports per unit
C	not that either, but the category has a default	quantity × the default — **flagged as an assumption**
D	nothing usable	applies the fallback and **flags it for review**

On the *Summary* sheet each rule appears with the letter **and** its meaning. If the line says 60 ports but the catalogue would give a different number for that quantity, the app **uses the explicit TEAP value** and records the discrepancy in the space's notes (the `Notes` column on Demand per space): that is where TEAP and catalogue disagree.

"Positions outside the page"

If the panel shows that warning with a table (`Kind` , `Label` , `Page` , `X` , `Y` , `Page size`), the drawing holds coordinates outside the sheet — which happens when a PDF with an unusual page rotation was imported by an older reader. The items are marked where they are stored, not moved — a marker that vanishes from an export with no explanation is worse than one visibly in the wrong place — but the fix is to re-import the VDB.

STEP 9 · AN OPTION

Composite map

The detail sheets of one level of one book laid on one map, in a window of its own, with everything already worked on the sheets — every NER and its reach, the Wi-Fi points, the TER, IDF, COW and DIVA markers. A sheet is the continuation of its neighbour; on one map the NERs of a level, their reach and the gaps between them read at a glance. It is built only when you ask for it, and the Drawing viewer is not touched by it.

What goes on a map

- **One map per level of one book.** The sheets are grouped by the level part of their drawing number (`412` in `VVV-412A1-XXX-OP`), or by the level in the title block when the drawing number does not say it. Two series are never mixed: a project with the 400 and the 600 books has maps of the 400 book and maps of the 600 book.
- **Never the key plan.** Key plans (ticked on the Scale screen) are not drawn: the map is laid out *against* them, as the reference, and is made of detail sheets only.
- **Only sheets with a scale.** A sheet with no scale is listed under its group as *without a scale, left out*.
- A level drawn on **one sheet** has nothing to lay out and gets no map.

How the sheets are placed

Every sheet is placed by two independent measurements, and the two check each other:

- **Match lines.** The heavy dashed lines a sheet prints where it continues, read against the grid the key plan draws with the same lines and the sheet's zone (`A1` in `412A1`). Exact when the book follows that pattern.
- **Correlation with the key plan.** The linework of the sheet compared with the key plan around the label the key plan prints for it (`412A1-XXX-OP` , or the zone letter). Works on

any book whose key plan shows the linework.

When the two disagree by more than 2 m the correlation is kept — the match lines assume a regular grid, and the books that break it are the ones where they disagree. Then every pair of sheets that overlap is compared directly, within 4 m of where they were put: a pair that lines up confirms both sheets and corrects a few metres at most; a sheet is never moved farther than that by its neighbours.

SHEET STATE	WHAT IT MEANS
`confirmed · blue outline`	The two measurements agree within 1 m, or a neighbouring sheet lines up with it, or a space code it shares with a neighbour lands in the same place.
`not confirmed · orange outline`	Placed by one measurement, with nothing to confirm it. Usually right; check it against its neighbours before deciding on it.
`contradicted · red outline`	Two or more space codes it shares with a neighbouring sheet do not line up: the sheet is probably in the wrong place. Leave it off the map.
`not placed`	Neither measurement could place it (no match line and no label for it on the key plan). Listed, never drawn.

Building and opening

On the Reports screen, the **Composite map** card lists every book and its levels.

CONTROL	WHAT IT DOES
`Build the maps of this book`	Places the sheets of every level of the book and composes one picture per level. It runs in the background; a book of sixty sheets takes a minute or two. Building again replaces the maps of the book and keeps the sheets you left off.
`Build`	The same, for one level only.
`Open ↗`	Opens the map in a window of its own. Several maps of the project open in the same window: choose another level in the list at the top.
`Delete`	Deletes the map — the picture, nothing else. Nothing on the sheets changes.

The map is a picture of the book: it is not in a backup and not in a project package. On another computer, or after a book is replaced (the card says *book replaced — build again*), build it again.

The map window

Scroll to zoom around the pointer, drag the background to move. The picture comes in pieces as you zoom, sharp at every scale; on top of it the application draws, as on the sheets:

- **Every NER** in its colour with its name, and its reach: the general limit as a dashed circle and the photographers' limit as a dotted one, straight line, to the map's scale. A NER drawn on two neighbouring sheets is one NER on the map; when its two drawings do not land in one place, both are shown, joined by a dashed line.

- **The Wi-Fi points** as light-blue dots, and the **TER, IDF, COW and DIVA** markers as small squares with their names.
- **The outline of every sheet** in the colour of its state, with its drawing number. `Spaces` shows the space codes when zoomed in.

Right after a map is built, or a sheet is left off it, the application draws in the background the scales the map opens at and the first steps in, so it opens already drawn. Closer scales are drawn as you reach them, a fraction of a second per piece, and are kept for the next time.

The boxes at the top turn each layer on and off. On the right:

PART	WHAT IT DOES
`Findings`	What only one plane shows: a NER drawn on two sheets whose drawings land more than 1 m apart (*drawn 10.2 m apart on two sheets*); two different NERs less than 2 m apart (*the same NER under two names?*); a NER drawn only on sheets left off the map. Click one to select the NER and centre the map on it.
`Sheets`	Every sheet of the level with its state. Click the drawing number to centre the map on it; `Open` opens it in the Drawing viewer of the main window; `Leave off` takes it off the map (it is not drawn and its NERs are not shown) and `Put back` returns it.
`Build this map again`	Places the sheets of this level again, after the scales or the key plan changed.

Deciding on the map

ACTION	WHAT HAPPENS
`Drag a NER`	Before anything changes, the question says **how far** the NER goes, in metres, and which sheets it **moves on**, is **taken off** or is **drawn on** — at the zoom that shows the whole level a short drag is a long way. After you confirm, the NER moves on **every sheet of the level that shows the new spot**; a sheet that no longer shows it loses that drawing of the NER; and when none of the sheets it was drawn on shows the spot, it is drawn on the sheet that does. Its drawings in another book are not moved (the notice names the book). Every change is in the audit log as a move, and the analysis is out of date until it runs again. A drawing taken off a sheet leaves its spot to the NER: `Place NERs from the drawing` reads the room printed there as this NER — it creates no new NER there and does not draw this one back on that sheet — not even where the sheet prints the NER's code somewhere else. A NER whose drawings land **more than 1 m apart** on the map (the *drawn … apart* finding: its sheets disagree, one of them is probably misplaced) is **not moved**: the notice names the sheets — check them, leave the misplaced one off the map, or move the NER on each sheet in the Drawing viewer. When the two drawings are on the **same sheet**, the notice says the NER is drawn twice there: delete the extra one with `Delete only this marker` in the Drawing viewer, then move it. A locked NER does not move.
`Undo`	In the notice after a move, and as `Undo the last move` on the right: every drawing of the NER goes back where it was, the ones taken off included. Refused when the NER was changed again after the move — move it back by hand. A sheet where the NER is drawn again at that spot is not given a second drawing of it; the notice names the sheet.
`Click a NER`	Selects it: its criticality, whether it is locked, the sheets it is drawn on and the other books.

ACTION	WHAT HAPPENS
Make critical / Make non critical	The same as changing the kind in the Drawing viewer: on the NER, on every drawing of it, confirmed.
Delete NER	The same as deleting it in the Drawing viewer: the NER leaves the register — every sheet, every book, the NERs list and the analysis. `Restore` on the NERs screen brings it back.
Open in the Drawing viewer	One button per sheet the NER is drawn on: the main window opens that sheet centred on it.

WHAT A MOVE CHANGES

Building a map, leaving a sheet off, putting it back and deleting a map change no number: they are pictures of the book. **Moving a NER is not a picture**: it changes where the NER is drawn on the sheets, so after the next analysis its distances, the statuses of the spaces it serves and the workbook change; until then the analysis is marked out of date (*changed since the last run*) and an export runs it again first. Every distance in the analysis is still measured on its own sheet, to that sheet's scale; the circles on the map are straight-line reach for the eye. A sheet placed by the application can be off by some decimetres when confirmed, and by more when not — decide on the orange ones only after checking them against their neighbours, and on a red one (contradicted) not before it is checked.

Composite map PDF

`Composite map PDF`, beside `Marked-up PDF` on the Exports card, writes one page per map built — the sheets as placed, each outlined by its state so that it reads without the colour — *confirmed*: blue dashed line; *not confirmed*: orange dotted line with **NOT CONFIRMED** beside the sheet number; *contradicted*: thicker red dash-dot line with **CONTRADICTED** — every NER with its name and its two reach circles, the Wi-Fi points and the other markers — with the legend — NERs, the two reaches, Wi-Fi, TER, IDF, COW, DIVA and the three sheet states — on the first page only (`with legend` off: no legend). A map larger than 200 inches is scaled to fit, the largest a PDF page may be. It does not replace the marked-up PDF, which stays the document of each sheet with the coverage.

ACCOUNT & SECURITY

Account & security

Sign-in accounts, administration, the audit log, and the statement of where the data lives.

The first time

On an installation with no accounts, the sign-in screen offers **Create the first account**: username, display name, password and the password again. That account is automatically an administrator. Projects already on the machine transfer to it. After that, open creation closes: new accounts come from an administrator only. **There is no default user and no default password** — if there were, it would be the same in every copy of the package.

Password: at least 10 characters, not beginning or ending with a space, and no upper-case-digit-symbol rule. Length is what resists guessing; composition rules push everyone toward `Passw0rd!` and buy very little. A phrase you remember is both stronger and easier to type every morning.

What the account changes day to day

- The whole API is closed until you sign in — the interface shows only the sign-in screen, with the data statement and the version.
- The audit log names **the person**. Every rename, classification, consolidation, choice of room and match is attributed.
- Projects are born **private**. Sharing is an explicit choice, and it grants reading, not editing — nor exporting: building a report counts as a write.
- A session lasts 12 hours. `Sign out` in the top bar ends it immediately.

Where it lives

At the end of the left-hand navigation, *below a divider*, under the heading `Administration` (or `Your account`, if you are not an administrator) — the tab reads `Accounts & security` for an administrator and `Account & security` for everyone else — or by clicking **your name in the top bar**. Cards, top to bottom:

CARD	WHAT IT DOES
`Your account`	Who is signed in, and changing your own password (`Current password`, `New password`, `Change password`). A red notice appears when the password was set by an administrator and should be changed.
`Add an account` ADMIN	Username, display name, initial password and the `administrator` box. The person signs in with the initial password and the app **asks them to change it**: a red notice on `Your account` and *change your password* in the top bar until they do. It does not block use — check afterwards that they changed it, so you do not end up knowing anyone's final password.
`Accounts` ADMIN	Table: `Username`, `Name`, `Projects` (which projects each one owns), `Last signed in`, open `Sessions`, the `Admin` and `Active` boxes, and the `Reset password` and `Delete` buttons. The boxes and buttons on your own row are disabled.
`Audit log`	Of the open project; with no project open there is nothing to show. `Load the audit log` fetches the 200 most recent entries. Columns: `When`, `Actor`, `Action`, `Entity`, `Justification`, `Payload`. `Actor` is the account signed in when the change was made, taken from the sign-in and never from what the page sends; `system` is a repair the application made by itself at start-up. It records every change: rename, classification, consolidation, removal, choice of room, match, scale, export.

CARD	WHAT IT DOES
Account events ADMIN	The account events — sign-ins, accounts created, blocked, reset and deleted, with the reason given — belong to no project, so they are not in a project's log. `Load the account events` fetches them. Administrators only: it names every account.
Where this data lives	The data statement, below. The same one appears on the sign-in screen.

What an administrator can do

ACTION	WHERE AND HOW
See every project	Automatic. On the `Projects` tab an administrator also sees other people's private ones, with the `Owner` column saying whose they are, and can edit them.
Block someone	Untick the `Active` box on their row. They can no longer sign in, and **any sessions they already had end immediately**. Their projects stay untouched. Ticking it again restores access.
Reset someone's password	`Reset password` on the row. You type the new one, and it is **flagged for change**, with the same notice at their next sign-in.
Promote or demote	The `Admin` box on the row.
Delete the account and its projects	`Delete` opens a red card listing **every project that goes with it** — venue, drawings, TEAP files and scenarios. It requires a **reason** (written to the audit log) and that you **type the username**; only then does `Delete permanently` unlock. No recycle bin.

BLOCK OR DELETE?

When in doubt, **block**. Disabling cuts access at once and preserves everything — one click reverses it. Deleting takes the projects with it and does not come back. When somebody leaves the programme, the normal move is to disable, export what matters, and only then decide whether the history is still worth anything.

WHAT THE APP REFUSES, EVEN TO AN ADMINISTRATOR

Disabling your own account; removing your own administrator rights; deleting your own account; and removing the **last active administrator**. An installation with no administrator cannot be recovered from inside the application — only by touching the database.

Forgot the password

On the sign-in screen, `Forgot your password?` says the two ways back in. The app has no e-mail and runs only on this computer, so neither goes through the browser:

- **Ask an administrator** of the installation: `Reset password` on your row under Account & security. You sign in with what they typed and the app asks you to choose your own.
- **No administrator — or you are the administrator?** On the computer where the app is installed, run the reset tool in the app's folder: `reset-password.bat` on Windows

(`%LOCALAPPDATA%\Programs\ner-analyzer\reset-password.bat` , double-click it) or `mac/reset-password.command` on a Mac (`~/Applications/ner-analyzer/mac/reset-password.command` , double-click it or run it in Terminal). It lists the accounts of the installation, asks which one and the new password twice (nothing is shown while you type; at least 10 characters), and saves it. The account is enabled again if it was disabled, every open session of it is signed out, and the reset is written to the *Account events* as *password reset on this computer*. It works with the app running; sign in with the new password.

The tool opens no new door: whoever can run a program in the installation folder can already copy the database. It is the owner of the computer's way back in, and the way out of an installation whose only administrator forgot the password.

The data statement

It appears on the sign-in screen and on this tab. It states **what the application does**, not which policy it complies with — every line is verifiable from outside the app:

- listens on `127.0.0.1` only, with no port on any network — another machine on the same network cannot reach it;
- makes no outbound connections — no project data goes anywhere, to any service, at any time;
- stores everything in one file on this machine, whose path the statement itself shows, and the uploaded documents as they were uploaded, under `data\uploads` ;
- records every change with actor, time and justification;
- stores passwords as scrypt hashes, never as text.

And what it says it does not protect: there is no encryption at rest — the database is protected exactly as far as this machine's disk and Windows account are, and anyone with file-system access can read it; sharing a project makes it readable by the other accounts *on this installation*, and publishes it nowhere; deleting a user deletes their projects, permanently. That is stated deliberately: a notice that only lists reassurances manufactures confidence that is not there.

Leaving this machine is always a manual act by a person exporting a file. The application never does it on its own, and any such release should follow your organisation's formal approval and cybersecurity process.

> IF THIS EVER BECOMES MULTI-USER ON A NETWORK
>
> This app's strongest protection today is not the password — it is that it listens on `127.0.0.1` only. Removing that so colleagues can reach it from other machines is a *different deployment* with different obligations: TLS, an owner for the server, backups, patching, access review — and this statement has to be rewritten to describe the server rather than the machine.

The distance rules

- A network point, device or access point sits at most **90 m** from its serving NER.
- A photographer point or area sits at most **70 m**.
- Both limits, the near-limit ratio (*Near-limit threshold*) and the slack factor are configurable per project in step 1.
- Where a cable route is traced, the distance is the route length. Where it is not, it is the straight line multiplied by the slack factor, and the result carries the warning *Preliminary straight-line distance. The result must be validated against the real cable route.*
- Cross-floor service is **not** assumed: it is only measured when a shaft, riser or vertical route is registered. The exception is your choice (or the TEAP's): a NER you chose on another page serves and counts against its capacity, with the distance reported as *not measured* until the vertical route exists — see Demand per space.
- A TER, an IDF, a COW or a DIVA node never serves. Where a room polygon exists, compliance is evaluated against its **farthest vertex**, not only its centre; where there is only a point, the point is the space's estimated centre and the analysis says so in the assumptions.

> READ BEFORE TRUSTING A GREEN RESULT
>
> A straight line is **not** cable length. Every straight-line result in this application is a screening number. The 90 m in ANSI/TIA-568 and ISO/IEC 11801 is the *permanent link*: installed cable along its real route, inside a 100 m channel that reserves 10 m for patch cords. Cable rises into a shaft, runs in tray around structure and drops back down, so a straight-line measurement systematically under-reports the installed length.
>
> Raise the `Routing slack factor` to 1.25–1.40 for an arena bowl, or trace the real containment, before treating a green result as a design decision. At 1.0 the app is reporting raw geometry, and says so in the notes column.

Every calculation is deterministic: the same input always produces the same result, and no language model takes part in any of it.

The coverage statuses

Four groups, which are what the screen and the workbook show — and, within each, the precise reason, which is what the engine computes.

From the desk, ten statuses are four situations: the space is served, it is served but violates something, it is served with no distance measured, or it is **not resolved** and needs a room on the drawing or a NER. So the pill in the `Status` column, the tiles and the filter of Demand per space, the workbook's `Status` column and the *Coverage figures* sheet speak of the **group**; the precise status stays on the small line under the pill, in the workbook's `Status detail` column and in the filter, because *match not confirmed* and *scale not confirmed* call for different actions.

GROUP	MEANS	REASONS IT GATHERS
Served	Measured, within the limit.	*within the limit · near the limit*
Out of limit / over capacity	Measured or counted, and failing: the distance exceeds the limit, or the room is out of ports.	*out of limit · insufficient capacity*
Served — not measured	A NER was chosen and the ports count against it, but no distance could be measured.	*other level — not measured · no room on the drawing — NER chosen by you*
Pending — needs a room or a NER	Not resolved. Tie the space to a room on TEAP × drawing, confirm what is missing, or choose the NER under `Served by`.	*no room on the drawing · not analysed yet · match not confirmed · scale not confirmed · route not validated · no NER reaches it*

The precise reasons

STATUS	MEANS
Served	Inside the limit.
Near the limit	Inside, but past the `Near-limit threshold`.
Out of limit	The distance exceeds the applicable limit — or no NER is within reach.
Insufficient capacity	Every NER within reach is out of free ports — or the NER you chose (or the TEAP named, or the one chosen for a space with no room) is out of ports; the room was assigned anyway and the overload shows on its load. It overrides *Match not confirmed* and *Scale not confirmed*, never *Out of limit*.
Match not confirmed	Only in an analysis run before version 3.0.0. Since then a space whose TEAP × drawing match is not confirmed is measured and served like any other, and the row says *room not confirmed* instead (see Demand per space). Run the analysis again and the status goes away.
Scale not confirmed	The page has a scale nobody confirmed — no definitive distance. A fixed NER still takes the space and its ports. A page with **no scale at all** does not land here: it measures nothing, and the space comes back *No NER reaches it*.
Route not validated	A measurement along a registered route was asked for and the route gave no length. In practice it does not appear: a straight-line distance gets its normal status (*Served*, *Out of limit*…) plus the *straight line* warning.
Other level — not measured	Served by a NER you (or the TEAP) chose on another page/level; the ports count against its capacity, and the distance was not measured because no shaft, riser or vertical route is registered between the two pages. Register the vertical route in the viewer to measure it.

STATUS	MEANS
`No room on the drawing — NER chosen by you`	The TEAP space is not on the drawing and you chose the NER that serves it; the ports count against its capacity and the distance was not measured, because there is no position to measure from. See A TEAP space the drawing does not show.
`No NER reaches it`	No NER is measurable from the space — none on the page and no vertical route, or the page has no scale at all — and nobody chose the NER under `Served by`. A wireless point whose space has no NER yet has it too. In the workbook and the API, `not_analysable`.
`Not analysed yet`	The space has a room on the drawing but no analysis has run since — press `Run analysis`.
`No room on the drawing`	The TEAP space was not found on any sheet (or you picked `— none —` on TEAP × drawing) and nobody chose the NER. The ports are on nobody's load.

Visual notation

The viewer legend and the exported-PDF legend list the same ten items in the same order (the first ten rows below). The last four rows — the space fill colours — are in neither legend; their meaning is in the `Status` column.

SYMBOL	MEANS
TER	Recorded and drawn, never serves.
NER — critical	A NER classified as critical.
NER — non critical	A non-critical NER. Its *name* is written in a darker shade than its dot: yellow is a good fill and bad ink.
IDF / MDF	Drawn and nothing else — never serves a space.
COW / temporary	A temporary cabin. Drawn and nothing else.
DIVA node	The DIVA rack. Read off the VDB or placed by hand; drawn and nothing else.
Wi-Fi — space with wireless	A Wi-Fi point, `WIFI-n`: the space has wireless. Placed from the TEAP (one per space) or by hand with `+Wi-Fi`. Not an access-point count, not a room and no port of its own: its TEAP space's `DWISA` line is the port. On the PDF the row reads *Wi-Fi - space with wireless*. Measured to the NER of its space so the 90 m rule is checked where it is drawn. On the marked-up PDF a disc in the status colour (served, near the limit, out of limit) sits under the sign.
Reference (key plan)	A dot in the room's colour with a **dashed ring**: a reference drawn on a key plan with `⋆NER★` / `⋆NER` / `⋆TER` / `⋆COW` / `⋆DIVA`, named after the real room. Drawn only: counted nowhere. A NER reference carries the reach circles to the sheet's scale; the others do not.
Named area	A slate-blue polygon with a faint fill and the name on a tag: an area drawn with `+area` to point something out (BTOC, TER, a compound). Illustrative only: nothing is measured or counted from it.

SYMBOL	MEANS
Match not confirmed	An orange rectangle around a space: the TEAP × drawing match for that space has not been validated by a person.
Area served	Area inside its limit.
Area near the limit	Inside, but past the `Near-limit threshold`. The same orange fills a space whose match is unconfirmed (which also gets the orange rectangle).
Area out of coverage	Outside the limit, or out of capacity.
Grey area	Not measured: scale unconfirmed, route not validated, or another level with no vertical route. A darker grey: no NER reaches the space.

Every marker is drawn **filled in its own colour with a hairline black ring**, on screen and in the exported PDF. A design book is dense black linework on white, and a filled dot in yellow or blue on top of it does not read as a separate object; the ring separates the marker from what it lands on without touching what the colour means. The viewer also shows: traced routes in teal, the general-limit circle in blue and the photographer one in purple, and the dashed amber ring of an unconfirmed marker.

> COLOUR DOES NOT CLASSIFY
>
> This convention is how the application **draws its own output** — it is not how it reads the source drawing. On real design books *every* equipment-room label tends to be the same red: `Critical NER 1`, `NER 5`, `COW #1`. Criticality is in the *text*. Never classify on colour alone.

Interface language

English by default, Portuguese by choice — and the line, which matters more than the button, between what is screen and what is data.

In the top bar, next to your name, there is a button marked `PT-BR` (or `EN` once you are in Portuguese). One click switches, at once, **without reloading the page**: the open project, the tab and anything half-typed stay where they were. The choice is kept in this browser on this machine. The same button sits on the sign-in screen.

TRANSLATED	NEVER TRANSLATED
Navigation, headings, buttons, field labels, table headers *on screen*, help text, empty states, the legend, the statuses, and the server messages you meet on a normal day (wrong password, disabled account, required field).	Space names, NER names and codes, venue codes, page labels, file names, the drawings, **anything that came out of the database**, the acronyms (NER, TER, TEAP, VDB, IDF, COW, DIVA) — and **every export**: XLSX, CSV and marked-up PDF come out in English whoever exported them, so the same report always reaches its reader in the same shape.

A space called `Observer Office 2` has to be found by searching `Observer Office 2`, whatever language the buttons are in: the name is what is written on the drawing, in the

TEAP and in everything that leaves here, and translating it would break the matching, the search and the traceability. Internal diagnostic messages stay in English on purpose: they are the ones quoted when something breaks.

Sorting the tables

Click any data column's heading. It works on every table in the application.

CLICK	WHAT IT DOES
1st click	ascending — the arrow becomes ▲
2nd click	descending — ▼
3rd click	back to the screen's **original order**

The third click exists on purpose: several lists arrive ordered for a reason — the TEAP × drawing queue by port demand, the audit log by time. **Blanks always sort last**, in both directions: “unknown” is neither a small value nor a large one. **Numbers inside text compare as numbers**: `NER-0002` before `NER-0010`, `p. 9` before `p. 74`. Columns that are actions rather than data (`Lock`, `Confirm`, `Details`, `Delete`) do not sort. Sorting is a view: it changes nothing in the database and does not affect the exports.

What the app does not do

Knowing this is part of using the tool responsibly.

LIMITATION	WHAT IT MEANS IN PRACTICE
Distances are not measured across sheets	The composite map lays the sheets of a level on one map, but it is a picture to decide on: distance is still only measured **within one page**. A space that appears on a sheet with a NER is measured there (see The same space on more than one page). But a space that exists **only** on a sheet with no NER, with the NER on the neighbouring sheet of the same level, comes back *No NER reaches it* — the way out is to choose the room under `Served by`: the choice counts the ports and reports the distance as not measured. The overall level plans cannot act as a common reference: they print no space codes.
Wired connection with no port in the TEAP	The TEAP counts the port on the device, not on the `Wired Connection` line. Where users bring their own equipment (photographers, press, commentators) nobody counts the port, and the application **does not invent it**: the space stays visible at zero ports with the warning, the distance is measured, and the NER's load **does not include** those connections. Treat the port total of those rooms as a floor, not a final figure. Wi-Fi is different: the TEAP requests a space's wireless as a line with **one** port (the access point's uplink), and that port always counts on the space — the access points drawn on the sheet are positions, never ports (see Demand per space).
A room with no text label does not exist for the app	A space is only read from a **code** or a **name label** in text on the sheet. A room drawn with no label, a label that is part of the image (a scanned plan) or a pavilion interior drawn as one block never becomes a space — the TEAP row stays unmatched and it is for you to pick the room on the worklist, or to accept that the book does not show it. The app creates no room where it reads nothing.

LIMITATION	WHAT IT MEANS IN PRACTICE
A NER read off the drawing is a reading, not a design	The placing step creates a NER where the drawing prints `TEC.010` / `TEC.020` (or `TEC.003.01` / `.02`), the *Network Equipment Rack* label or the NER block symbol, and it takes part in the analysis until it is deleted. On a book the team has already commented (the red NERs), the architect's rooms more than 12 m from every red NER become new NERs — perhaps rooms the team decided not to use. Check the `from the room label` rows before carrying numbers forward. The arrow is only followed when it is a thin line to a solid arrowhead; any other shape (a curve, no head) leaves the NER on the label, with the warning. A label with no code on two sheets becomes two NERs — consolidate them on the NERs screen. The block symbol is recognised against one reference block (one design team's); a venue drawn with a different block — another word, another lettering — is not read until its block is added to the reference. A room drawn with no text and no symbol is not read.
TVDB is not read	Containment, pathway, Route A/B, shafts and redundancy are **not analysed**. Every distance with no traced route is a straight-line screening number, and every NER carries “no TVDB route validated” as a pending validation.
A traced route and a measure are flat	The `route` and `measure` tools measure **on the plan**, on the sheet's one scale: they do not add the rise from the room to the tray, the drop to the outlet or the rack, or the service loops — all part of the 90 m permanent link. A route of 85 m with two 4 m drops is already over. And a page has one scale: a line drawn over an enlarged detail printed at another scale on the same sheet is measured at the wrong one. Read both as the horizontal part of the cable, not the installed length.
No venue guard in the matcher	A space from another venue can match your drawing. The only defence is the venue filter in step 1. **Always check** that matched spaces belong to the project's venue.
No drawing versioning	A second VDB *version* of the same drawings in one project is read as more sheets of the same venue: the pages do not collide, but the old and the new rooms are measured side by side as though both were built. There is no reconciliation screen — the app cannot answer “did they draw what I asked?”. Use a new project per version.
A project package is a copy, not a link	Importing a package creates an independent project. Changes made on either copy never reach the other, and there is no merge: agree who carries on before anyone works on the copy. A package written by a newer version of the app cannot be imported by an older one, and it is **not encrypted** — it holds every design book of the project.
No OCR	Scanned pages are *flagged*, not processed. They show `OCR needed` and nothing is extracted from them.
Spaces are points	The room polygon is not extracted from the drawing; the space is estimated as a point. Large rooms carry more error.
Level comes only from the title block	A sheet's level is read from the title block's *DRAWING TITLE* (`KEY ENLARGED PLAN - LEVEL 01` → `LEVEL 01`). A sheet whose title does not state the level has none, and the duplicate queue asks instead of merging. The level the *drawing number* implies (`-440B5-` → level 04) is used only for the “same space on another page?” question, as weaker evidence: good for ruling another floor out and for making your answer easy, never for linking by itself.
One machine, one writer at a time	It runs locally and listens on `127.0.0.1` only. Several accounts, but no concurrency control: sharing grants reading, and only the owner (or an administrator) writes.
No two-point calibration	There is no tool to click two points on the sheet and type the distance between them. A page's scale comes from the printed scale, the title block or the ratio you type on the Scale screen; a sheet with no printed scale and no known ratio cannot be measured.

LIMITATION	WHAT IT MEANS IN PRACTICE
Jobs do not survive a restart	An import, a matching run, an analysis or a PDF export that is running when the server stops is lost, not resumed. Run it again once the server is back; nothing half-written is kept.
No scenario restore	Every analysis replaces the project's one As-Is scenario. An earlier result survives only as the workbook you exported from it.
No automatic NER proposal	There is one proposal, and it is the drawing: the placing step reads the team's annotations and the rooms and symbols the design team already drew (`TEC.010` / `TEC.020` , `TEC.003.01/02` , the *Network Equipment Rack* label, the NER block), but it invents no position — new NERs are placed by hand in the Drawing viewer. The tool's job is to say what each placement costs — the distances, what falls outside the limits, what nothing reaches. Power, cooling, physical space, security and A/B independence are not in the model, and that is why the decision is yours.

Running and maintenance

How it starts

After `setup.bat` , by itself: a Windows scheduled task starts the server at every sign-in, with no window, and the **NER Analyzer** desktop shortcut opens it in the browser. By hand, two ways. `start.bat` opens a console window that *is* the server — closing the window stops the application — and opens the browser at `http://localhost:8000` as soon as it answers. Or, after running `autostart.bat` once, a Windows scheduled task named **NER Spatial Coverage Analyzer** starts the server at sign-in, with no window, and `stop.bat` stops it.

```
task     .python\pythonw.exe  tools\serve.py   ·   trigger: at sign-in
log       data\server.log
manual    start.bat   ·   stop.bat   ·   tools\server-start.cmd   ·   tools\server-stop.cmd
```

> COLD START: 20 TO 40 SECONDS
>
> Loading the libraries and running the database migration takes that long. Allow that margin before concluding it failed. The line `Application startup complete` in `data\server.log` marks the moment it was actually ready.

Memory

With the libraries loaded the server sits at about 350 MB. Reading a book — the import, and `Place NERs from the drawing` — happens in a separate Python process that exits when the job ends, so the server does not grow with the books it has read. The browser is the other half: the Drawing viewer keeps one book open at a time and closes it when you change drawing, project or screen. If the machine is short of memory all the same (a laptop with many other things open), the import and the viewer are the first to slow down, well before anything fails; closing other programs helps more than restarting the server.

The watchdog: it comes back on its own

The second task `autostart.bat` creates, **NER Analyzer watchdog**, runs every two minutes, probes `/api/auth/state` **twice** (one timeout during a heavy export is not a dead server) and starts the server again if it does not answer — it never restarts one that is answering. Nor one that is *starting* (a process younger than three minutes), nor one that is merely *slow* (the port accepts the connection but no answer comes within 20 s): that one is restarted only after three consecutive checks, six minutes. One check runs at a time — the next waits for the previous one to finish. Before restarting it writes the machine's memory state, the five largest consumers and the standby and power events of the last half hour to `data\watchdog.log`, so the cause of a fall is recorded rather than reconstructed later. After the start it polls the health check for up to 150 s. `data\watchdog-heartbeat.txt` is a single line, rewritten on every check: that is how to know the watchdog is alive without waiting for a failure.

> A LAPTOP ON BATTERY
>
> A task created by Windows is born with three defaults that kill a server silently: *do not start on battery*, *stop when switching to battery* and *end after 72 hours*. `autostart.bat` turns all three off on both tasks (`tools\task-harden.ps1`) and says, for each, `[ok] … runs on battery, is not stopped on battery, has no time limit`. If it says `[*]`, the task still works on mains power, and the message names the boxes to untick in Task Scheduler (the *Conditions* and *Settings* tabs). On an installation older than this version, run `autostart.bat` once.
>
> **How to tell the watchdog is alive:** open `data\watchdog-heartbeat.txt`. The time in it must be no more than two or three minutes old. A stale time means the watchdog stopped running — and that a fallen server will not come back by itself.
>
> **In the browser, while the server is out:** the top bar shows the red pill *server not answering — reconnecting*, every action answers *the server is not answering…*, and the page polls `/healthz` every five seconds. When the server is back, the session and the project are reloaded by themselves and the notice says *the server is back*. A server that is down does not end the session; the sign-in screen appears only when the 12-hour session has actually expired.
>
> `tools\server-start.cmd` starts the server **through the scheduled task** when it exists, and only without it as a process of its own. The difference matters: a process started by a session (a remote terminal, a tool) dies with it; started by the Task Scheduler it belongs to nobody's session.

Backup

Everything lives in `data\ner.sqlite3` (plus the `-wal` and `-shm` files) and in `data\uploads\`. `backup.bat` writes a consistent copy of the database — valid even while the server is running, because it uses SQLite's backup API instead of copying the file — and the uploaded documents, into `data\backup-YYYYMMDD-HHMMSS\` (or a `backup-…` folder inside the folder you name with `backup.bat --to D:\folder`).
`backup.bat --restore data\backup-…` puts a backup back, with the server stopped; what was there is kept in `data\replaced-YYYYMMDD-HHMMSS\`, not deleted. A backup can be restored into another folder or on another computer with nothing to edit: when the server next starts, it points the record of every drawing and TEAP at this installation's own

`data\uploads`. A backup before deleting a project or installing a new version costs seconds.

When something does not respond

1. Wait two minutes — with the watchdog installed, it brings the server back by itself. If it does not, look at the time in `data\watchdog-heartbeat.txt`: stale, the watchdog is not running — start the server with `tools\server-start.cmd` and run `autostart.bat` again.
2. Read the end of `data\server.log`. If the server is up and the page is blank or odd, press `Ctrl` + `Shift` + `R` — the browser is holding the old interface; the amber pill in the top bar says exactly when that is happening.
3. `stop.bat` then `start.bat`; read `data\watchdog.log` to find out why it fell. A fall with no traceback and no `===== stopped =====` line in the log is a process killed from outside — memory, leaving standby or a policy of the machine; the watchdog log records the memory and the power events of the previous half hour. Close the browser and Outlook before exporting a whole book.
4. Run `.python\python.exe tools\check_env.py` — it names any missing dependency instead of throwing a traceback mid-request.
5. Port 8000 taken by another program: `start.bat --port 8080`. The watchdog always probes port 8000: a server on another port is not watched.

Command-line tools

COMMAND	WHAT FOR
`run-tests.bat`	Runs the application's test suite on this installation (around 900 tests, one to two minutes). All passing, the installation is sound.
`tools\check_env.py`	Lists the dependencies present and the ones missing.
`tools\check_title_blocks.py [max pages]`	Shows, per book under `data\uploads`, the first pages (6 by default; the optional number changes it) with the drawing number, title and level the title-block reader finds. Run it on a book from a new family before trusting the numbers.
`tools\validate_against.py <vdb.pdf> <teap.xlsx> --venue VEE`	Extraction report for a pair of documents: what was extracted, which scales, which demand rules fired, how much the matching resolved and how much was held for review. All local; nothing is uploaded.
`build_package.bat`	Builds this version's installation package into `dist\` — see the installation manual.

The Python scripts run with the installation's private Python, which lives inside the application folder: `.python\python.exe tools\<script>.py`, from that folder. (On a development checkout without the embedded Python, the path is `.venv\Scripts\python.exe`.)

Installing on another machine

To move **one project** — not the application — to another computer, use `Export` and *Import a project package* on the Projects screen.

One .zip file, a few clicks at the other end, no administrator rights needed, and no internet connection. The full walkthrough — prerequisites, installation, first sign-in, updating, backup, uninstalling and common problems — is in the **installation manual**, `docs\INSTALL.html`, which ships with the package.

In short: on the source machine, `build_package.bat` writes `dist\ner-analyzer-5.2.2-windows.zip` with the application, the compiled screens, the manuals, the tests, every Python package and Python itself — and **no data**: no projects, no drawings, no accounts. On the target machine there is nothing to install first: unzip anywhere, right-click `setup.bat` → *Run as administrator* (a double-click asks Windows for the rights itself) — or, where administrator rights cannot be granted, double-click `setup-noadmin.bat`, which installs for that account only — and create the first account. `setup.bat` installs into `%LOCALAPPDATA%\Programs\ner-analyzer` for the person signed in, registers the sign-in task and the watchdog, starts the server, opens the browser, puts an **NER Analyzer** shortcut on the desktop and deletes the installation files, leaving only the application. An existing installation is backed up and replaced: its data is kept, and the old version is moved aside to `ner-analyzer.previous`. Opening `install.bat` in the package hands over to `setup.bat`.

On a Mac (Apple Silicon — M1 or later — macOS 11 or later) the package is `ner-analyzer-5.2.2-macos-arm64.zip`, built with `python tools/make_package.py --platform mac`. It carries its own Python 3.12 and the same pinned packages as the Windows one, and needs no administrator password: `setup.command` installs into `~/Applications/ner-analyzer`, registers a launch agent for that account (the server starts at sign-in and again if it stops — what the watchdog does on Windows), starts the server, opens the browser, puts an **NER Analyzer** link on the Desktop and removes the installation files. An existing installation is backed up and replaced, its `data` kept. The `mac` folder of the installation holds `start`, `stop`, `backup`, `run-tests`, `autostart` and `uninstall`, each a `.command`. The steps, and what to do when macOS says it cannot verify the developer, are in the installation manual.

Version and licence

This manual describes **version 5.2.2** of the NER Spatial Coverage Analyzer. The installed version appears on the sign-in screen, in the sidebar under the title, at `http://localhost:8000/healthz`, on the *Summary* sheet of every exported workbook (*Generated by*) and on every marked-up PDF (at the end of the legend on the first page, and in the footnote of every other page). The `CHANGELOG.md` file that ships with the application says what changed in each version and how versions are numbered.

There is no artificial intelligence inside this application. Nothing in this folder calls out to any model, service or account; you can prove it by turning the network off and using the application as usual.

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